

Software Requirements Specification for Zainul Abedin VR Museum

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1. Introduction

The Zainul Abedin VR Museum offers a virtual experience that showcases the art and life of the renowned Bangladeshi painter Zainul Abedin, making it accessible to people around the world. Instead of visiting the physical museum in Mymensingh, visitors can explore a 3D version using their computer or phone, simulating an in-person visit.

Utilizing web technologies like WebXR and A-Frame, users can navigate through the galleries, examine paintings in detail, and enjoy audio guides available in multiple languages. They can also view Zainul Abedin's tools and personal belongings, enhancing the experience to be both engaging and educational.

This virtual museum serves as a valuable resource for students, researchers, and art enthusiasts looking to deepen their understanding of Zainul Abedin's life and his significant contributions to Bangladesh's art history. Our aim is to make his art easily accessible to everyone, ensuring that his work continues to inspire future generations.

1.1 Purpose of the Document

The purpose of this Software Requirements Specification (SRS) is to provide a detailed overview of the functional and non-functional requirements for the Zainul Abedin VR Museum. This document serves as a reference for developers, designers, testers, project managers, stakeholders, and future maintenance teams involved in the creation and operation of the system.

The SRS outlines the system's goals, features, behavior, and constraints to ensure a clear and mutual understanding among all parties. By documenting the requirements, this SRS helps to:

Provide a solid foundation for the design and development of the system.

Serve as a guideline for project planning and testing.

Act as a communication tool between technical and non-technical stakeholders.

1.2 Scope of the System

The Zainul Abedin VR Museum is a web-based application designed to offer an immersive virtual tour experience of the physical museum dedicated to the legendary Bangladeshi artist, Zainul Abedin. The system will use modern web technologies such as WebXR and A-Frame to deliver a 3D interactive environment accessible through desktop and mobile devices.

Key features of the system include:

- A virtual gallery that simulates the layout and exhibits of the real museum.
- Interactive viewing of Zainul Abedin's artworks with zoom and rotate functions.
- Multilingual audio guides providing commentary and historical context.
- Display of the artist's personal belongings and tools in 3D.
- Informational pop-ups and text panels for additional learning.
- User-friendly navigation and accessibility options for a broad audience.

1.3 Definitions, Acronyms, and Abbreviations

This section provides definitions for all document names, acronyms, and abbreviations. The application domain's terms and concepts are defined.

Term/Acronym	Definition
VR	Virtual Reality – a simulated experience that can mimic or differ from the real world.
WebXR	A web API that enables immersive experiences (VR and AR) through web browsers.
A-Frame	A web framework for building 3D/VR experiences using HTML and JavaScript.
3D	Three-dimensional – having height, width, and depth.
SRS	Software Requirements Specification – this document.
UI	User Interface – the visual part of the application that users interact with.
Zainul Abedin	A prominent Bangladeshi painter, known for his realistic and socially conscious artwork.
Gallery	A virtual room or area showcasing a specific group of artworks or exhibits.
Artifact	Any object or exhibit (e.g., paintings, tools, personal belongings) displayed in the virtual museum.
Audio Guide	A narrated commentary accompanying museum exhibits, available in different languages.

2. Overall Description

2.1 Product Perspective

The Zainul Abedin VR Museum is a **standalone web-based application** that simulates a virtual museum environment. It is not part of a larger system but may integrate with the following external components:

Web Browsers: The application relies on modern web browsers that support WebXR APIs (e.g., Chrome, Firefox, Edge).

Content Delivery Networks (CDNs): Used for hosting static assets such as 3D models, textures, audio files, and scripts.

Multilingual Audio Services: May integrate with third-party tools or content repositories to support language translations and audio playback.

Analytics Tools: Optional integration with analytics platforms (e.g., Google Analytics) to track user interactions for usability improvements.

The system is self-contained in terms of functionality and does not require database integration for core operation, though future versions may add user accounts or personalization features.

2.2 User Classes and Characteristics

User Class	Description	Experience Level
General Visitors	Casual users exploring the museum out of interest or curiosity.	Basic computer/mobile skills.
Students	School and university-level learners using the platform for educational purposes.	Moderate digital literacy.
Researchers	Art historians, scholars, and academicians studying Zainul Abedin's work in depth.	Advanced users, expect detailed information.
Educators	Teachers and lecturers who may use the platform as a teaching aid.	Moderate to advanced users.
Art Enthusiasts	Individuals with a keen interest in fine arts and virtual exhibitions.	Varies from novice to expert.
Museum Staff/Admin (Future Scope)	Personnel responsible for maintaining or updating the virtual museum.	Technical and content expertise.

2.3 Operating Environment

The system is expected to function in the following environments:

2.3.1 Hardware Requirements

- Desktop or laptop computer with modern GPU (recommended).
- Smartphone with gyroscope (for immersive experience).
- Optional VR headset (e.g., Oculus Quest, HTC Vive) for enhanced immersion.

2.3.2 Operating Systems

- Windows 10 or later
- macOS 11 or later
- Android 10 or later
- iOS 14 or later

2.3.3 Web Browsers (latest versions)

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari (limited WebXR support)

2.3.4 Network Requirements

- Internet connection of at least 5 Mbps for optimal performance.
- Hosting server capable of serving 3D content with low latency.

2.4 Design and Implementation Constraints

2.4.1 Technology Stack

- Must use **A-Frame** and **WebXR** for 3D and VR rendering.
- Frontend developed using **HTML5**, **JavaScript**, and **CSS3**.
- All assets (images, models, audio) must be optimized for web delivery.

2.4.2 Performance Constraints

- 3D models and textures must be optimized to reduce loading time.
- Audio files should be compressed without losing clarity.

2.4.3 Browser Compatibility

- Must function on all major modern browsers.
- Some advanced WebXR features may not be available on all platforms.

2.4.4 Accessibility Requirements

- Support for keyboard and mouse navigation.
- Audio and text-based alternatives for visually or hearing-impaired users.

2.4.5 Regulatory and Ethical Constraints

- Must comply with copyright and licensing laws for all media content.
- Multilingual content must be culturally accurate and respectful.
- GDPR compliance for any data tracking or user analytics (if implemented).

3. Functional Requirements

The functional requirements define the specific behaviors and functions the system must perform to fulfill the needs of the museum visitors and other stakeholders. These requirements ensure that users can easily navigate the virtual museum, interact with exhibits, and access information about the artworks and artefacts on display.

3.1 User Registration

FR-1	Users can create an account by registering with email and password to access personalized features.		
Description	New users provide email, password, and basic information to create an account and use features like saving favorites and receiving notifications.		
Stakeholders	Users	Priority	High
User Story-1	As a new user, I want to register using my email and password, so that I can create a personal account and access personalized features.		

3.2 Email Verification

FR-2	System verifies user's email before activating the account.		
Description	Upon registration, the system sends a verification email with a link. Users must verify to activate their account and access full features.		
Stakeholders	Users	Priority	High
User Story-2	As a registered user, I want to verify my email address via a verification link, so that I can activate my account and access all features.		

3.3 User Login

FR-3	Registered users can log into the system securely.		
Description	Users enter verified credentials to access personalized and restricted content.		
Stakeholders	Users	Priority	High
User Story-3	As a registered user, I want to log into the system using my credentials, so that I can securely access my personalized content.		

3.4 Password Recovery

FR-4	Users can recover or reset forgotten passwords.		
Description	Users can request a password reset link sent to their email to regain account access.		
Stakeholders	Users	Priority	High
User Story-4	As a user, I want to recover my forgotten password via email, so that I can regain access to my account.		

3.5 Virtual Museum Navigation

FR-5	User can explore the virtual museum in 3D view.		
Description	Users (general visitors, students, artists, or researchers) can enter and move around the museum using a browser. The system will render 3D environments with clickable artworks and navigation tools (keyboard-mouse).		
Stakeholders	Users	Priority	High
User Story-5	As a user, I want to navigate through the virtual museum in a 3D environment, so that I can explore exhibits as if I were physically there.		

3.6 Interactive Artwork Viewing

FR-6	Users can click artworks to view detailed information.		
Description	Users select artworks to see expanded descriptions, history, and artist notes in information panels.		
Stakeholders	Users	Priority	High
User Story-6	As a user, I want to click on artworks to view detailed information, so that I can learn more about their history and context.		

3.7 360-Degree Artwork View

FR-7	Users can view artworks with 360-degree rotation and zoom.		
Description	Select artworks allow users to rotate and zoom in for detailed examination in a virtual space.		
Stakeholders	Users	Priority	Medium

User Story-7	As a user, I want to rotate and zoom in on certain artworks, so that I can examine them from all angles for better appreciation.
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3.8 Search Functionality

FR-8	Users can search for artworks and exhibits by keywords.		
Description	The system provides search input with filters to help users find specific artworks, periods, or topics quickly.		
Stakeholders	Users	Priority	Medium
User Story-8	As a user, I want to search for artworks and exhibits using keywords and filters, so that I can find specific content quickly.		

3.9 Multilingual Support

FR-9	The system supports Bangla and English languages.		
Description	Users can toggle between Bangla and English for UI and content to suit their preference.		
Stakeholders	Users	Priority	Medium
User Story-9	As a user, I want to toggle between Bangla and English, so that I can navigate and understand the content in my preferred language.		

3.10 Save Favorite Artworks

FR-10	Users can bookmark artworks for quick access later.		
Description	Users add artworks to a favorites list accessible in their profile for future visits.		
Stakeholders	Users	Priority	Medium
User Story-10	As a user, I want to bookmark my favorite artworks, so that I can easily find and revisit them later.		

3.11 Audio Guide Integration

FR-11	Users can listen to audio guides in multiple languages.		
Description	Audio narration explains artworks and exhibitions as users navigate, available in Bangla and English.		
Stakeholders	Users	Priority	Medium
User Story-11	As a user, I want to listen to audio guides in my preferred language, so that I can better understand the exhibits while exploring.		

3.12 Admin Add/Edit Artworks

FR-12	Admins can manage the museum's digital collection.		
Description	Authorized personnel add, update, or remove artworks and exhibit details to keep content current.		
Stakeholders	Admin	Priority	Medium
User Story-12	As an admin, I want to add or edit artwork entries, so that the virtual museum content remains accurate and up to date.		

3.13 Admin User Management

FR-13	Admins manage user accounts and permissions.		
Description	Admins can activate, suspend, or assign roles to user accounts for security and management.		
Stakeholders	Admin	Priority	High
User Story-13	As an admin, I want to manage user accounts and permissions, so that I can ensure secure and appropriate access control.		

3.14 User Feedback Submission

FR-14	Users can submit feedback about their experience.		
Description	A form allows visitors to share comments, suggestions, or report issues, helping improve the virtual museum.		
Stakeholders	Users, Admin	Priority	Medium

User Story-14	As a user, I want to submit feedback or report issues, so that I can help improve the virtual museum experience.
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3.15 Exhibit Visit Tracking

FR-15	System tracks which exhibits a user has visited.		
Description	To enhance personalized experience, the system logs user interactions and optionally recommends unseen exhibits.		
Stakeholders	Users, Admin	Priority	Medium
User Story-15	As a user, I want the system to track which exhibits I have visited, so that I can receive recommendations and avoid repeat visits.		

3.16 Artefact Presentation

FR-16	The system shall display 3D models or images of artefacts related to Zainul Abedin.		
Description	The system will showcase personal belongings, tools, and other artefacts associated with Zainul Abedin, either as 3D models or high-quality images.		
Stakeholders	Users	Priority	High
User Story-16	As a user, I want to view Zainul Abedin's artefacts in 3D or high-quality images, so that I can better appreciate his personal and artistic legacy.		

3.17 Print & Download Feature for Educational Use

FR-17	Users can print or download exhibit summaries and educational materials.		
Description	Some exhibits offer downloadable PDFs or printable summaries for teachers, researchers, or offline use.		
Stakeholders	Students, Teachers, Researchers	Priority	Medium
User Story-17	As a teacher or researcher, I want to print or download educational materials, so that I can use them for teaching or offline study.		

3.18 Responsive Design

FR-18	The system shall be responsive and accessible on various devices (desktops, tablets, and smartphones).		
Description	The VR museum will adapt to different screen sizes and resolutions, providing a consistent experience across devices.		
Stakeholders	Users	Priority	High
User Story-18	As a user, I want the virtual museum to work on all devices (PC, tablet, phone), so that I can have a consistent experience anywhere.		

4. Non-Functional Requirements

The non-functional requirements define how the system should perform rather than what it should do. These characteristics ensure that the virtual museum delivers a smooth, secure, and high-quality experience across devices and use cases.

4.1 Performance Requirements

NFR-1	The system shall load the initial virtual museum environment within 5 seconds on a standard broadband connection (≥ 5 Mbps).		
Description	Optimized assets and asynchronous loading shall be used to reduce waiting times and improve user satisfaction.		
Stakeholders	Users	Priority	High

NFR-2	The system shall maintain an average frame rate of 30 FPS or higher on supported devices.		
Description	To ensure smooth 3D navigation, graphics will be optimized for desktop and mobile devices without compromising visual quality.		
Stakeholders	Users	Priority	High

NFR-3	Audio playback shall start within 1 second of user interaction.		
Description	Audio guides should load with minimal delay to support an immersive and educational experience.		
Stakeholders	Users	Priority	Medium

4.2 Usability Requirements

NFR-4	The system shall be intuitive and require no prior training for general users.		
Description	Menus, icons, and interactions will be self-explanatory and aligned with standard web practices.		
Stakeholders	Users	Priority	High

NFR-5	The interface shall support both Bangla and English, and users can switch languages at any point.		
Description	Multilingual support shall be consistent across UI, audio guides, and content.		
Stakeholders	Users	Priority	Medium

NFR-6	The system shall comply with WCAG 2.1 Level AA accessibility standards.		
Description	Support for keyboard navigation, screen readers, captions for audio, and high-contrast mode will be provided.		
Stakeholders	Users with disabilities	Priority	Medium

4.3 Reliability Requirements

NFR-7	The system shall maintain 99.5% uptime per month, excluding scheduled maintenance.		
Description	Hosting infrastructure will use load balancing and redundancy to ensure high reliability.		
Stakeholders	Users, Admin	Priority	High

NFR-8	The system shall auto-recover from temporary loading or asset failures without needing user intervention.		
Description	If a resource fails to load, the system will retry or offer a fallback option to maintain continuity.		
Stakeholders	Users	Priority	Medium

4.4 Availability Requirements

NFR-9	The system shall be accessible 24/7 from any location with internet access.		
Description	Cloud-based deployment ensures continuous availability across global time zones.		
Stakeholders	Users	Priority	High

NFR-10	The system shall notify users with a friendly message in case of downtime or maintenance.		
Description	Custom messages with estimated restoration times will be shown when the site is temporarily unavailable.		
Stakeholders	Users	Priority	Medium

4.5 Security Requirements

NFR-11	All user data and system communications shall be encrypted using HTTPS.		
Description	SSL/TLS certificates will be used to prevent eavesdropping and ensure data integrity.		
Stakeholders	Users, Admin	Priority	High

NFR-12	Passwords shall be stored using strong hashing algorithms (e.g., bcrypt).		
Description	User credentials will never be stored in plain text, ensuring data protection.		
Stakeholders	Users	Priority	High

NFR-13	The system shall comply with GDPR and other relevant privacy standards.		
Description	Users must be informed and opt-in for data tracking, cookies, or analytics.		
Stakeholders	Users	Priority	High

4.6 Maintainability Requirements

NFR-14	The system codebase shall be modular and use standardized components.		
Description	Component-based design (e.g., using A-Frame entities or Web Components) allows easier updates or replacement		
Stakeholders	Developers, Admin	Priority	Medium

NFR-15	Content (artwork data, audio files, metadata) shall be managed through structured formats like JSON or XML.		
Description	Admins can easily update exhibit content without modifying core system logic.		
Stakeholders	Admin	Priority	Medium

4.7 Portability Requirements

NFR-16	The system shall be compatible with all major browsers (Chrome, Firefox, Safari, Edge) and their mobile versions.		
Description	Cross-browser testing will ensure consistent functionality across platforms.		
Stakeholders	Users	Priority	High

NFR-17	The system shall gracefully degrade when used on low-end devices (e.g., disable advanced VR features).		
Description	Fallback modes (like 360° view instead of WebXR) will be provided for broader access.		
Stakeholders	Users	Priority	Medium

5. Requirement Prioritization

5.1 MoSCoW Method

ID	Requirement	Priority	Category	Reasoning
FR-1	User Registration	Must Have	M	Required for personal features like favorites and tracking.
FR-2	Email Verification	Must Have	M	Prevents spam and ensures valid users.
FR-3	User Login	Must Have	M	Core security and personalization feature.
FR-4	Password Recovery	Must Have	M	Essential for account access recovery.
FR-5	3D Virtual Museum Navigation	Must Have	M	Central to the virtual museum experience.
FR-6	Interactive Artwork Viewing	Must Have	M	Key engagement feature for understanding exhibits.
FR-7	360-Degree Artwork View	Should Have	S	Enhances immersion but not essential for all users.
FR-8	Search Functionality	Should Have	S	Improves usability, especially for large collections.
FR-9	Multilingual Support (Bangla/English)	Should Have	S	Increases accessibility for diverse audiences.
FR-10	Save Favorite Artworks	Could Have	C	Useful for repeat visits but not essential for first-time users.
FR-11	Audio Guide Integration	Should Have	S	Adds educational value; important for non-readers or visually impaired.
FR-12	Admin Add/Edit Artworks	Must Have	M	Ensures content remains current and manageable.
FR-13	Admin User Management	Must Have	M	Required for managing security and user access.
FR-14	User Feedback Submission	Could Have	C	Improves UX over time but not core functionality.
FR-15	Exhibit Visit Tracking	Could Have	C	Personalization feature; adds value but not mandatory.
FR-16	Artefact Presentation (3D or Image)	Must Have	M	Core to the museum's goal of showcasing Zainul Abedin's legacy.
FR-17	Print & Download for Education	Should Have	S	Valuable for students/teachers; adds educational utility.
FR-18	Responsive Design	Must Have	M	Critical for accessibility across all user devices.

5.2 Kano Model

ID	Requirement	Kano Category	Justification
FR-1	User Registration	B – Basic	Users expect to create an account for personalization.
FR-2	Email Verification	B – Basic	Standard for security and spam prevention; expected by users.
FR-3	User Login	B – Basic	Fundamental for returning user access.
FR-4	Password Recovery	B – Basic	Expected usability feature to prevent lockout.
FR-5	3D Virtual Museum Navigation	P – Performance	Core to system value; satisfaction increases with ease and quality of navigation.
FR-6	Interactive Artwork Viewing	P – Performance	Directly influences engagement and learning.
FR-7	360-Degree Artwork View	E – Excitement	Delightful but not expected; impresses users with immersive exploration.
FR-8	Search Functionality	P – Performance	Expected by regular users; improves usability with growing content.
FR-9	Multilingual Support	P – Performance	Boosts satisfaction among Bangla- and English-speaking users.
FR-10	Save Favorite Artworks	E – Excitement	Not essential but appreciated; adds personalized experience.
FR-11	Audio Guide Integration	P – Performance	Increases understanding and satisfaction, especially for educational users.
FR-12	Admin Add/Edit Artworks	B – Basic	Required for system maintenance; users assume content stays current.
FR-13	Admin User Management	B – Basic	Standard administrative control; critical for security and operation.
FR-14	User Feedback Submission	E – Excitement	Users feel heard and valued; enhances community interaction.
FR-15	Exhibit Visit Tracking	E – Excitement	Adds personalized recommendations and user memory; unexpected but engaging.
FR-16	Artefact Presentation (3D or Image)	P – Performance	Strong impact on the educational and immersive experience.
FR-17	Print & Download for Education	E – Excitement	Teachers/researchers appreciate it, but most users do not expect it.
FR-18	Responsive Design	B – Basic	Users expect compatibility across devices as a minimum requirement.

5.3 100-Point Model

ID	Requirement	Points	Priority Level
FR-5	3D Virtual Museum Navigation	12	Very High
FR-6	Interactive Artwork Viewing	10	Very High
FR-1	User Registration	8	High
FR-3	User Login	7	High
FR-16	Artefact Presentation	7	High
FR-2	Email Verification	6	High
FR-18	Responsive Design	6	High
FR-11	Audio Guide Integration	6	High
FR-9	Multilingual Support	5	Medium
FR-8	Search Functionality	5	Medium
FR-12	Admin Add/Edit Artworks	4	Medium
FR-4	Password Recovery	4	Medium
FR-13	Admin User Management	3	Medium
FR-7	360-Degree Artwork View	3	Low
FR-10	Save Favorite Artworks	2	Low
FR-14	User Feedback Submission	1	Low
FR-15	Exhibit Visit Tracking	1	Low
FR-17	Print & Download for Education	0	Very Low (Optional)

6. Validation and Verification Plan

Validation ensures the system meets user needs, while verification ensures the system is built correctly according to the specifications.

6.1 Validation Techniques

To confirm that the system satisfies stakeholder needs and expectations:

- **Stakeholder Walkthroughs**
Conduct structured walkthroughs of prototypes and early builds with end users, educators, and museum staff to validate alignment with expectations.
- **User Feedback Sessions**
Deploy beta versions to a sample audience (e.g., students, researchers, general users) and collect feedback through surveys and interviews.
- **Requirement Review Workshops**
Organize review meetings with stakeholders to confirm that all requirements are understood, necessary, and feasible.
- **Usability Testing**
Observe real users navigating the virtual museum to ensure the experience is intuitive, accessible, and educational.

6.2 Verification Techniques

To verify that the system functions as intended and meets the documented requirements:

- **Requirements Inspections**
Formal reviews of the SRS to detect inconsistencies, omissions, or ambiguities.
- **Unit and Integration Testing**
Test individual components (e.g., navigation, login) and then their integration to verify correct interactions.
- **System Testing**
Conduct end-to-end testing of the full system to confirm all requirements are met.

- **Traceability Analysis**

Maintain bidirectional traceability between requirements and implementation to ensure each one is fulfilled.

- **Automated Testing**

Use automated test scripts for regression, performance, and UI responsiveness checks across platforms.

6.3 Requirement Traceability Matrix (RTM)

The RTM ensures that each functional requirement is tested and can be traced throughout development.

Requirement ID	Requirement Description	Test Case ID(s)	Validation Method	Verification Method
FR-1	User Registration	TC-01, TC-02	Walkthrough, Feedback	Unit Test, System Test
FR-2	Email Verification	TC-03	User Feedback	System Test
FR-3	User Login	TC-04	Walkthrough	Unit Test, System Test
FR-4	Password Recovery	TC-05	Review, Feedback	Integration Test
FR-5	3D Virtual Navigation	TC-06, TC-07	Usability Testing	UI Test, Integration Test
FR-6	Interactive Artwork Viewing	TC-08	Walkthrough, Feedback	System Test
FR-7	360-Degree Artwork View	TC-09	User Review	Functional Test
FR-8	Search Functionality	TC-10	Stakeholder Demo	Unit Test, Integration Test
FR-9	Multilingual Support	TC-11	Language Testing	UI Test, Internationalization Test
FR-10	Save Favorite Artworks	TC-12	Usability Feedback	System Test
FR-11	Audio Guide Integration	TC-13	Walkthrough, Feedback	Integration and Audio Test
FR-12	Admin Add/Edit Artworks	TC-14	Admin Feedback	CRUD Operation Test
FR-13	Admin User Management	TC-15	Admin Review	Access Control Test
FR-14	User Feedback Submission	TC-16	Review & Feedback	System Test
FR-15	Exhibit Visit Tracking	TC-17	Beta Test Feedback	Data Logging Test
FR-16	Artefact Presentation (3D/Image)	TC-18	Usability Testing	3D Rendering Test
FR-17	Print & Download Educational Materials	TC-19	Teacher Review	Download Test
FR-18	Responsive Design	TC-20	Cross-device Testing	UI Test on Desktop & Mobile

7. UML Diagrams

7.1 Use Case Diagram

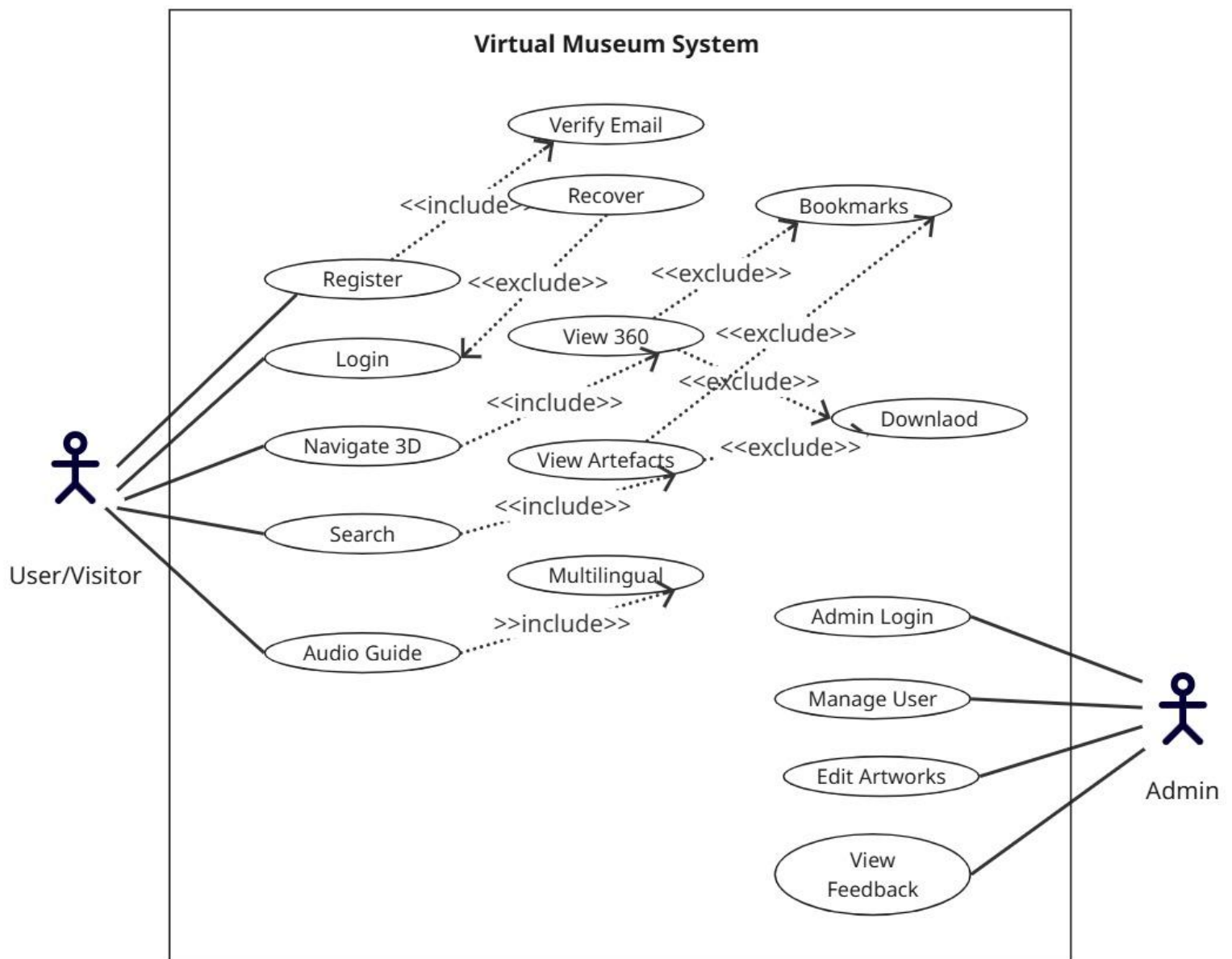


Figure 1: Use Case Diagram

7.2 Class Diagram

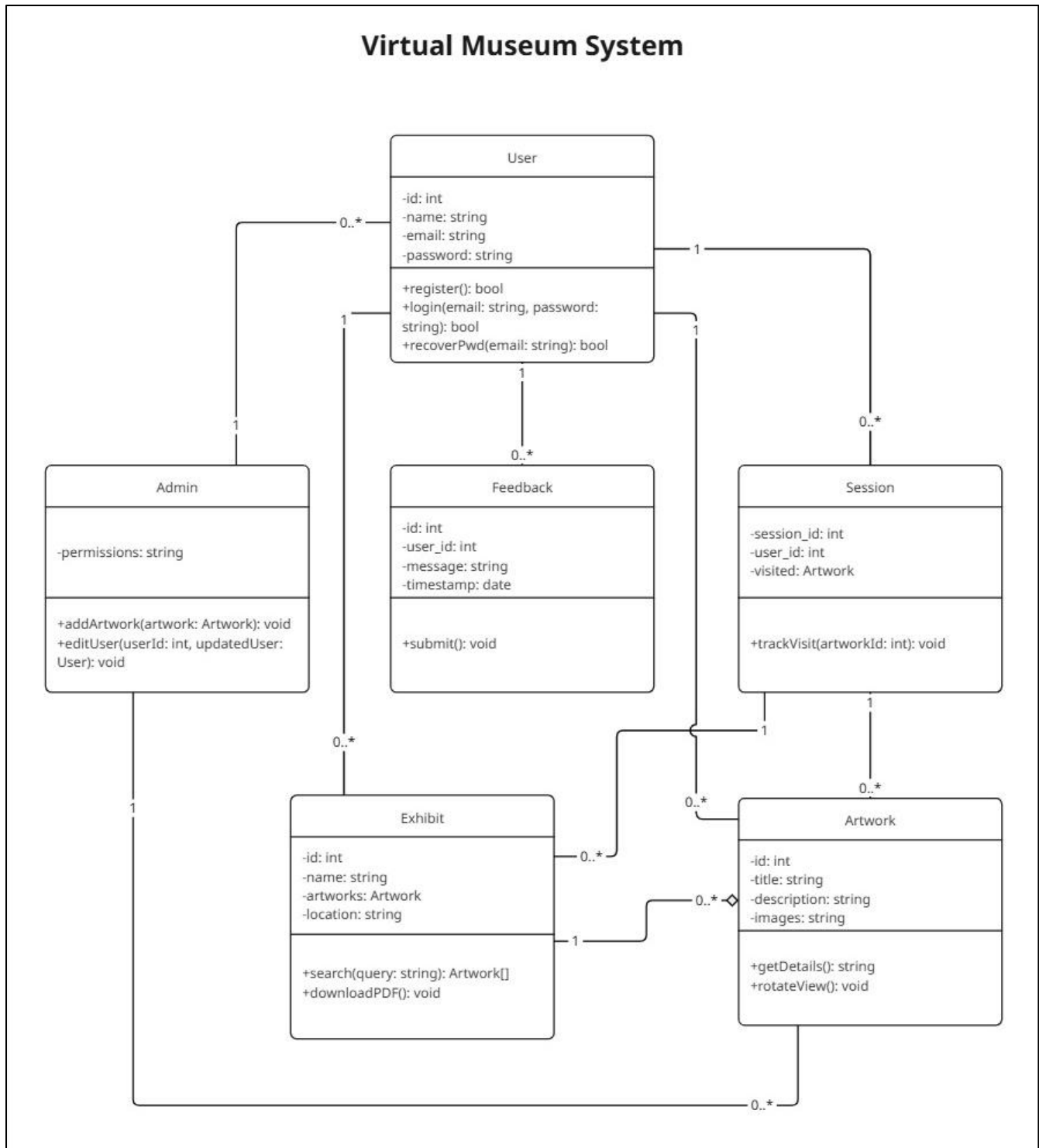


Figure 2: Class Diagram

7.3 Activity Diagram

7.3.1 Sign Up

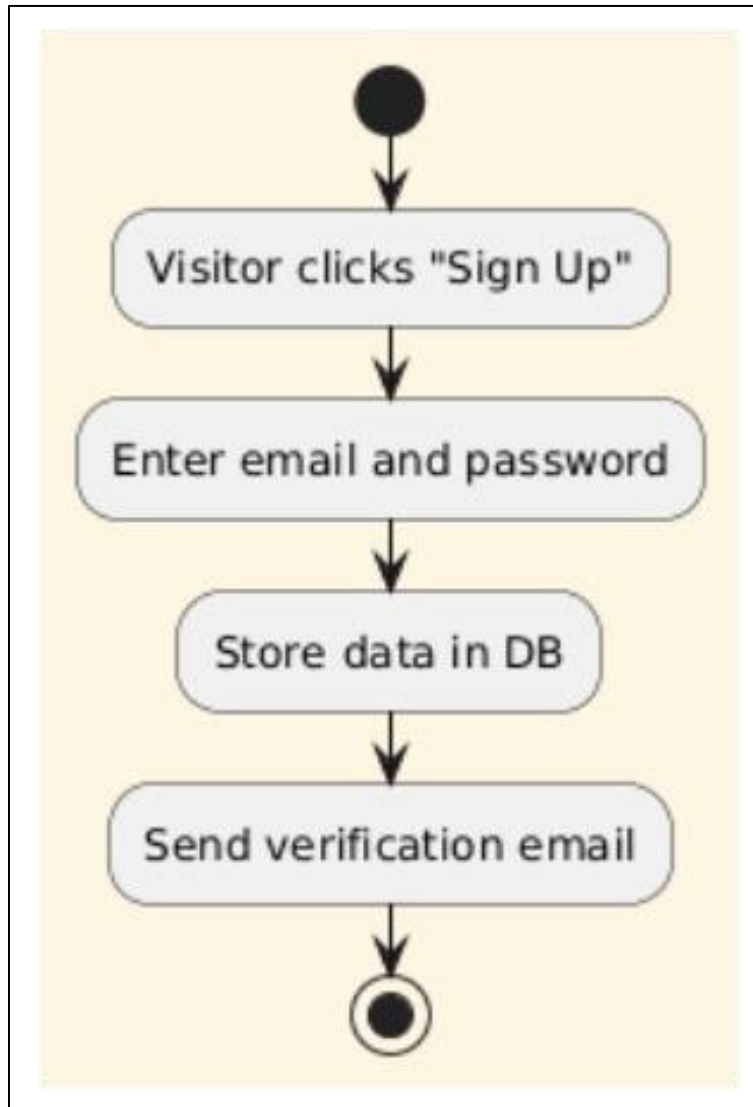


Figure 3: Sign Up

7.3.2 Email Verification

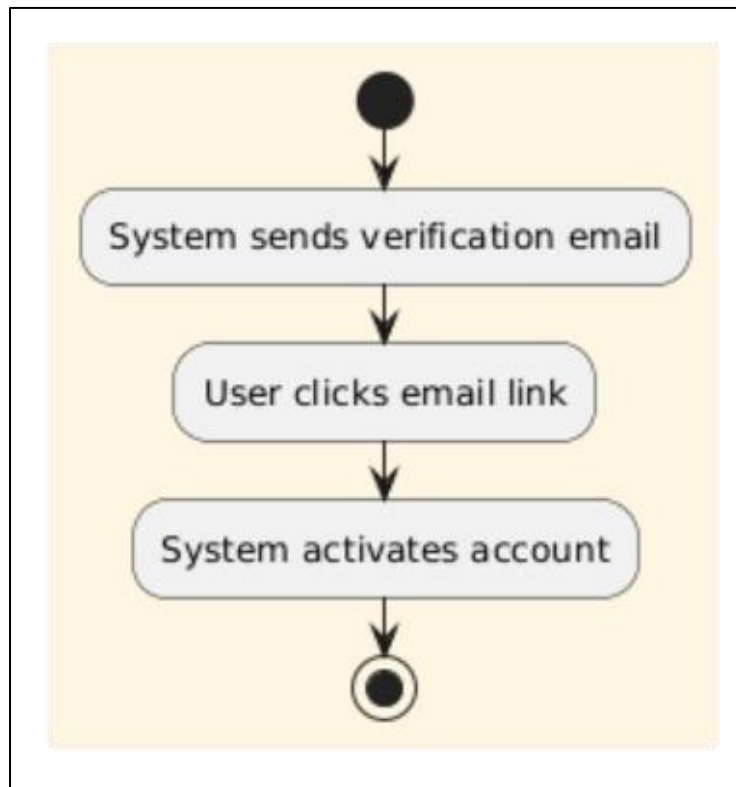


Figure 4: Email Verification

7.3.3 Login

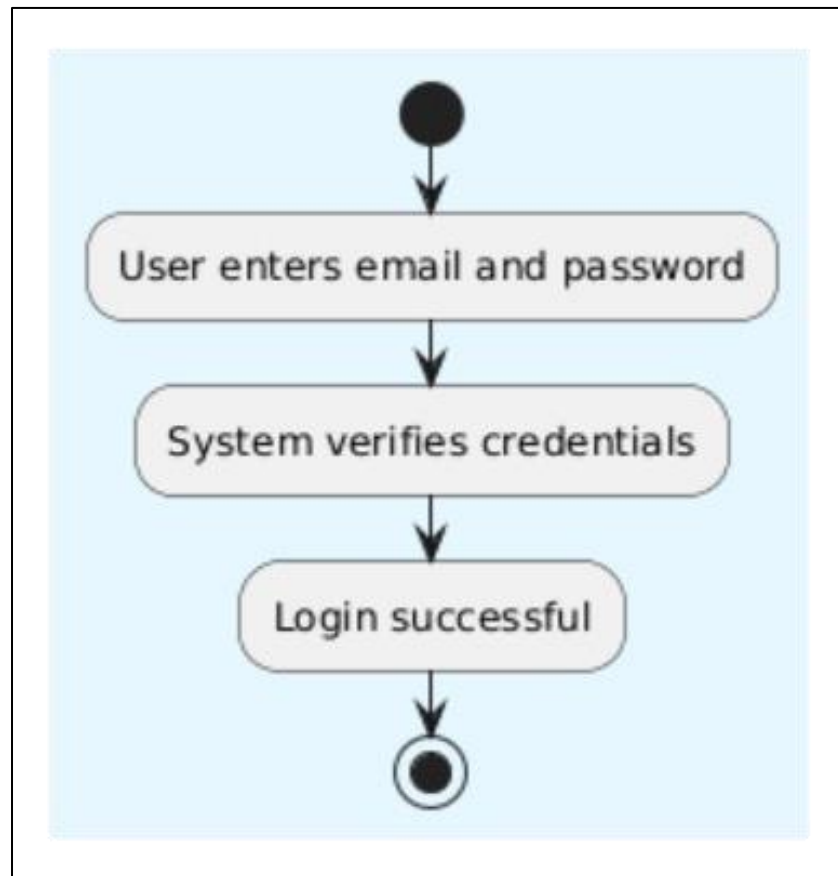


Figure 5: Login

7.3.4 Reset Password

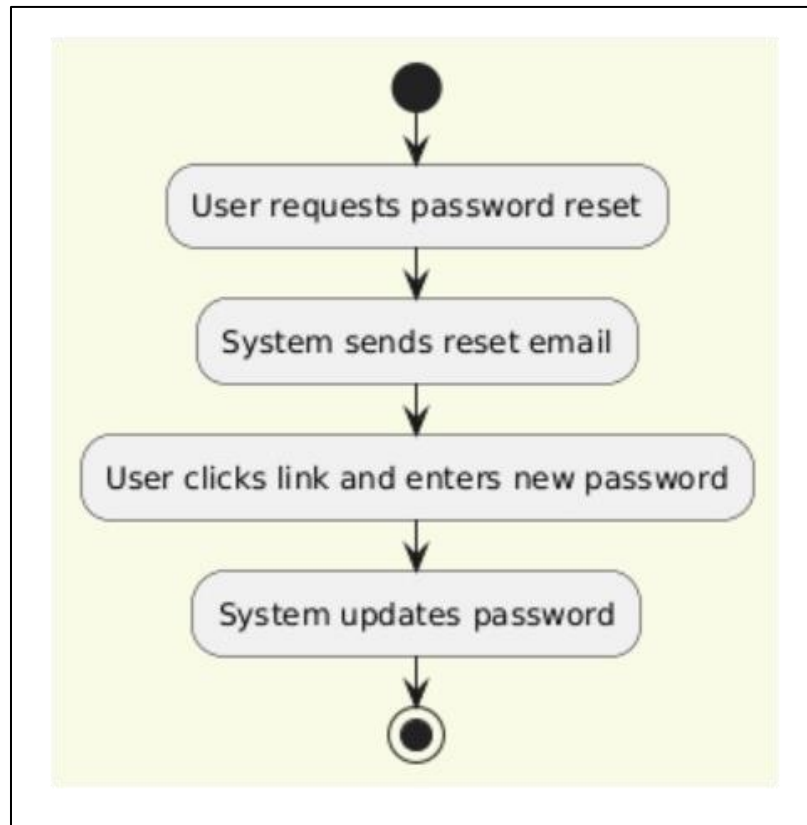


Figure 6: Reset Password

7.3.5 Navigate Museum

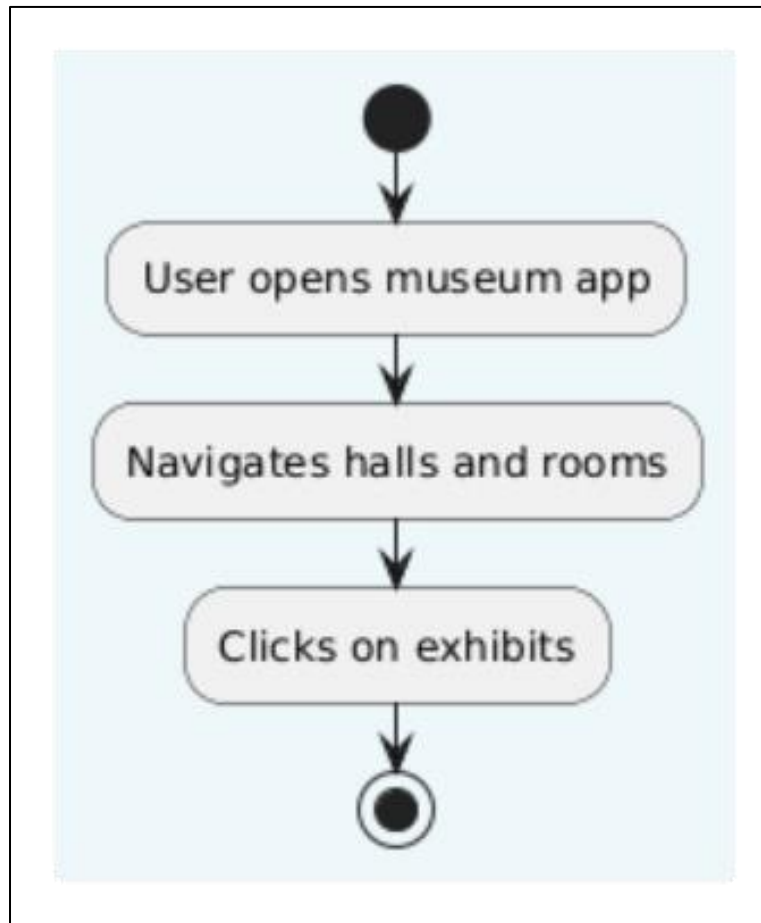


Figure 7: Navigate Museum

7.3.6 View Artworks

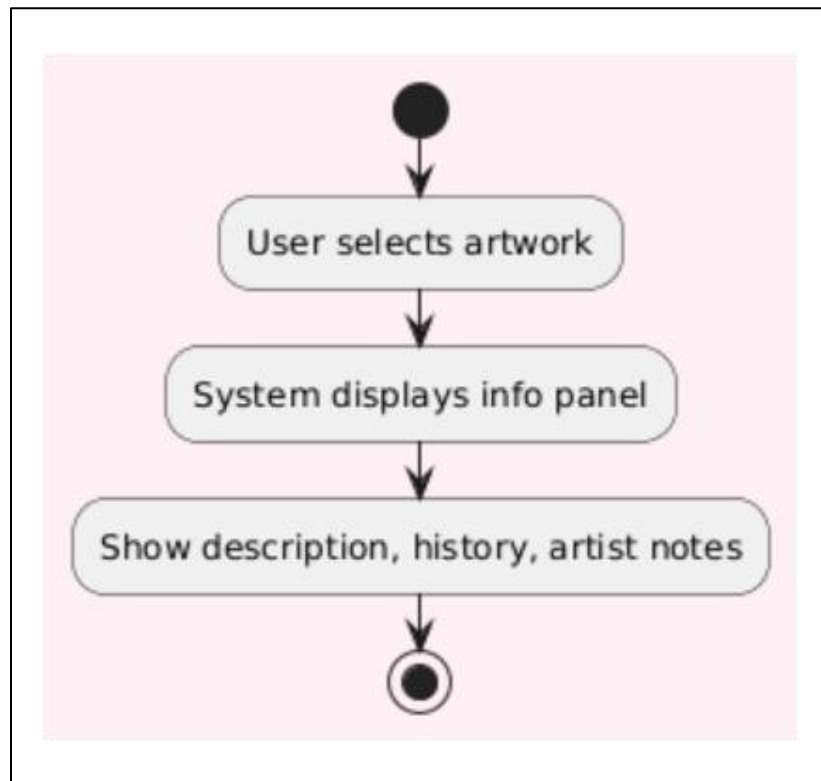


Figure 8: View Artworks

7.3.7 View 3D

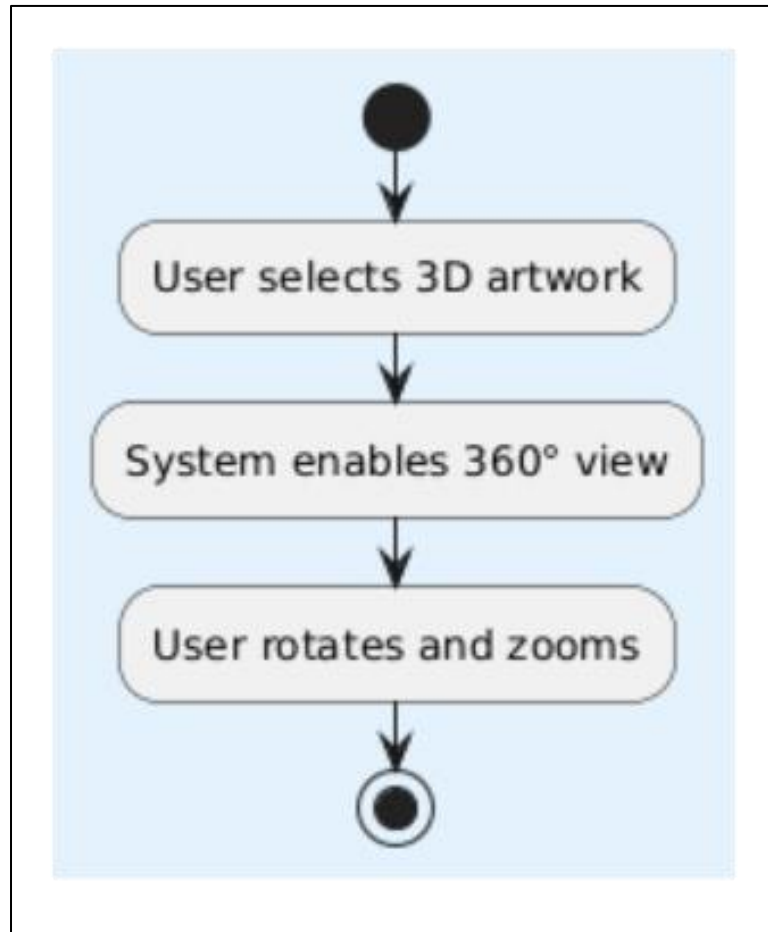


Figure 9: View 3D

7.3.8 Search

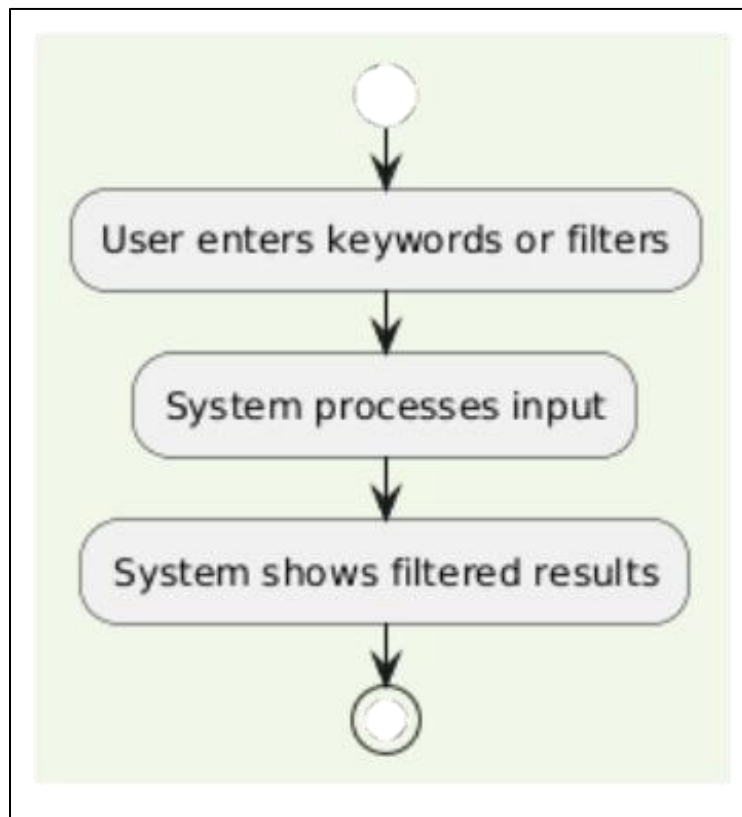


Figure 10: Search

7.3.9 Change Language

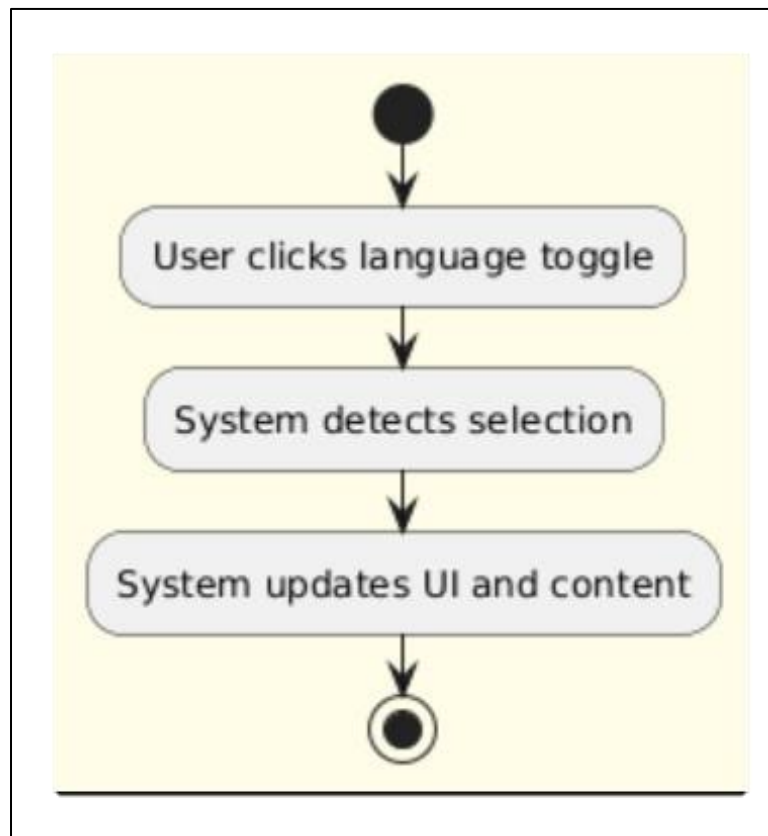


Figure 11: Change Language

7.3.10 Bookmark

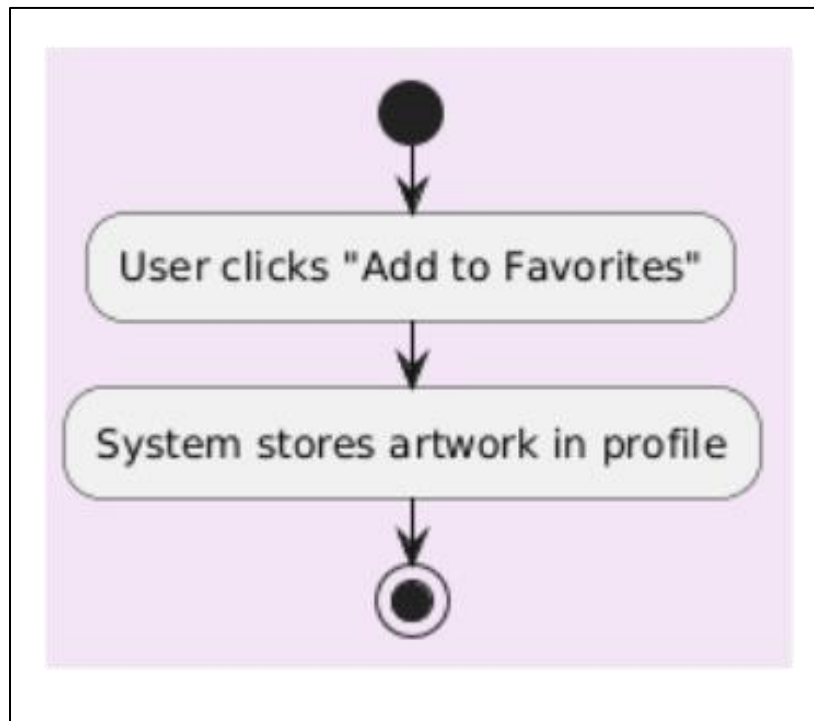


Figure 12: Bookmark

7.3.11 Enable Audio Direction

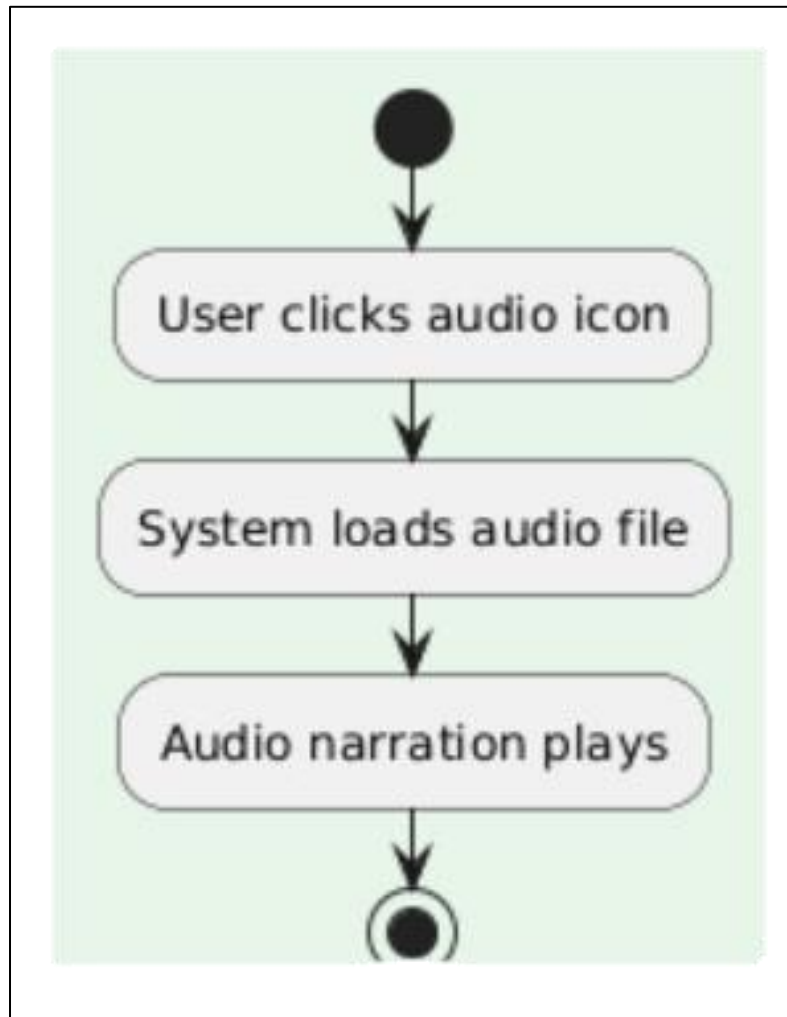


Figure 13: Enable Audio Direction

7.3.12 Edit Artworks

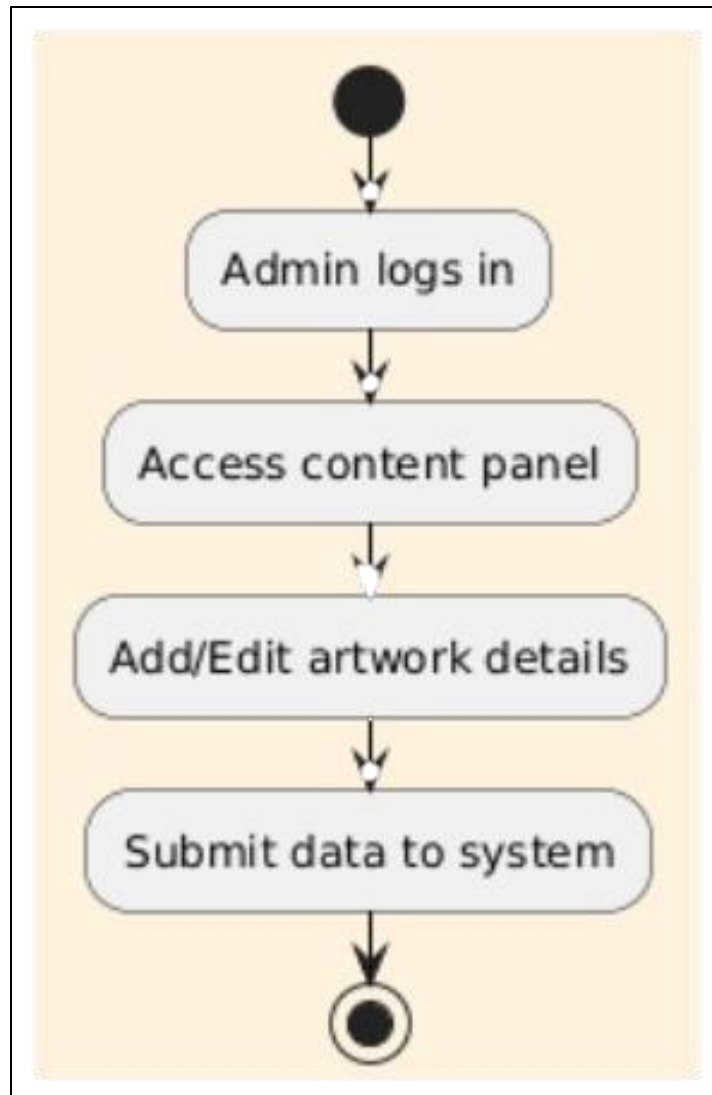


Figure 14: Edit Artworks

7.3.13 Manage User

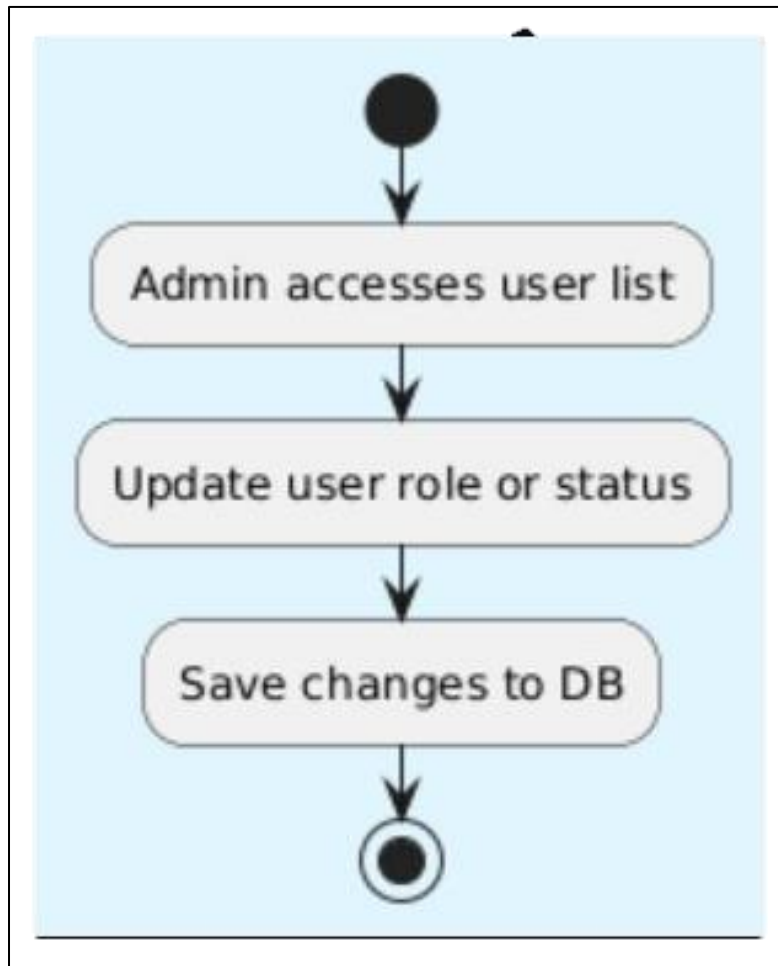


Figure 15: Manage User

7.3.14 Submit Feedback

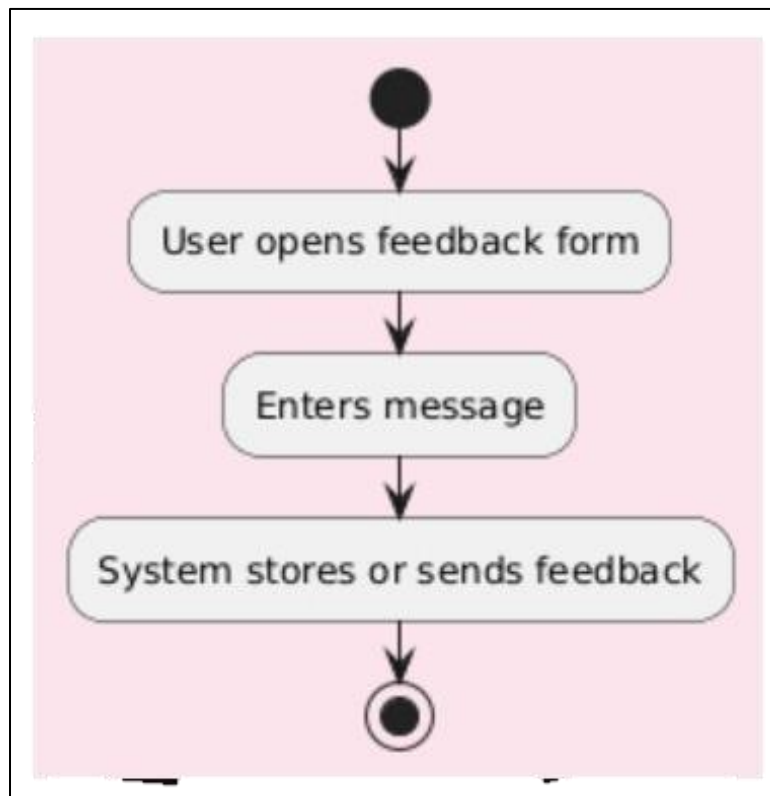


Figure 16: Submit Feedback

7.3.15 Visit Tracking

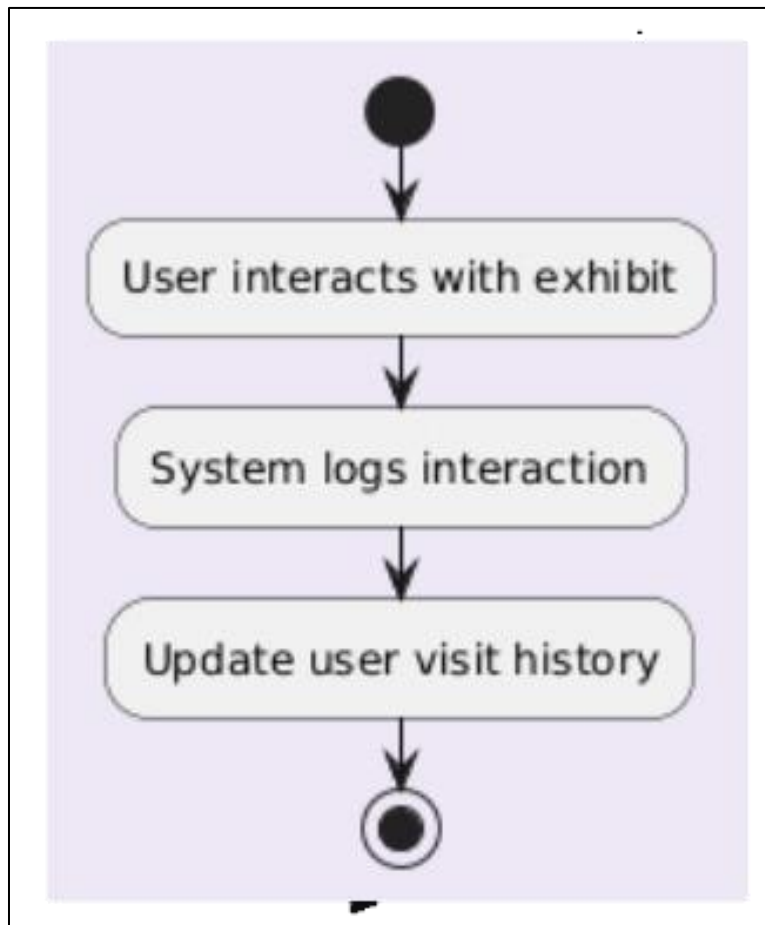


Figure 17: Visit Tracking

7.3.16 Artefacts Presentation

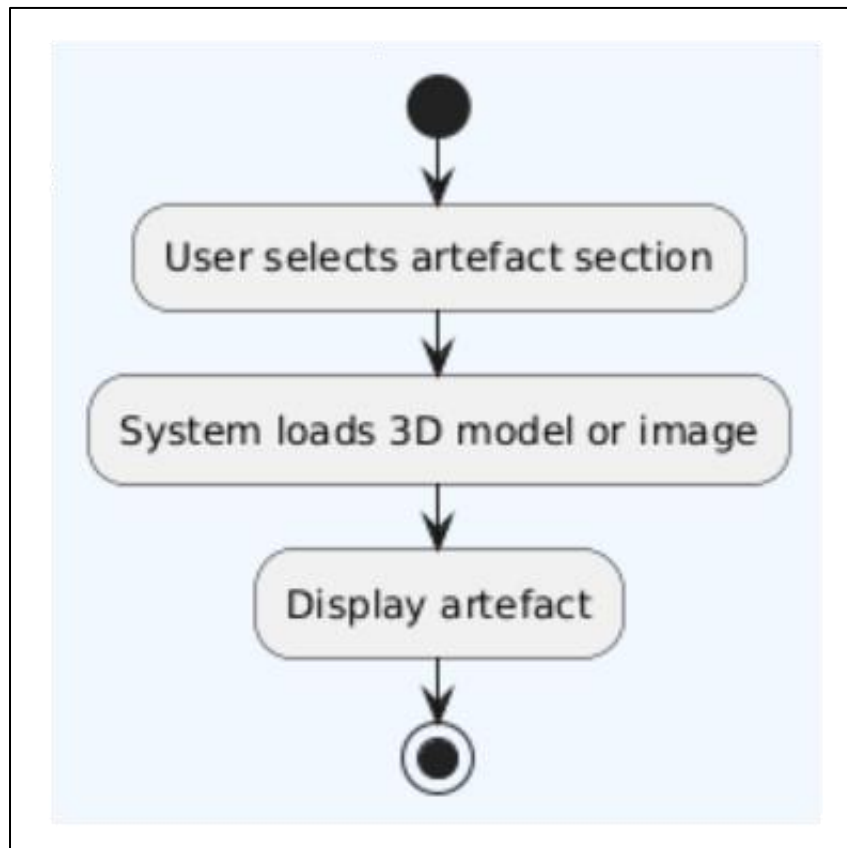


Figure 18: Artefacts Presentation

7.3.17 Print & Download

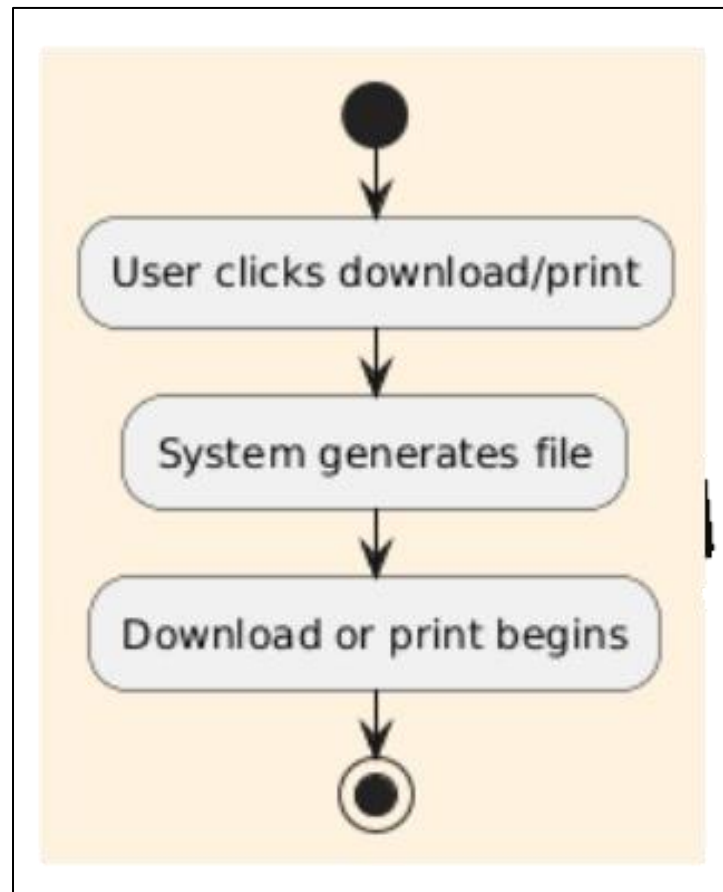


Figure 19: Print & Download

7.3.18 Responsive

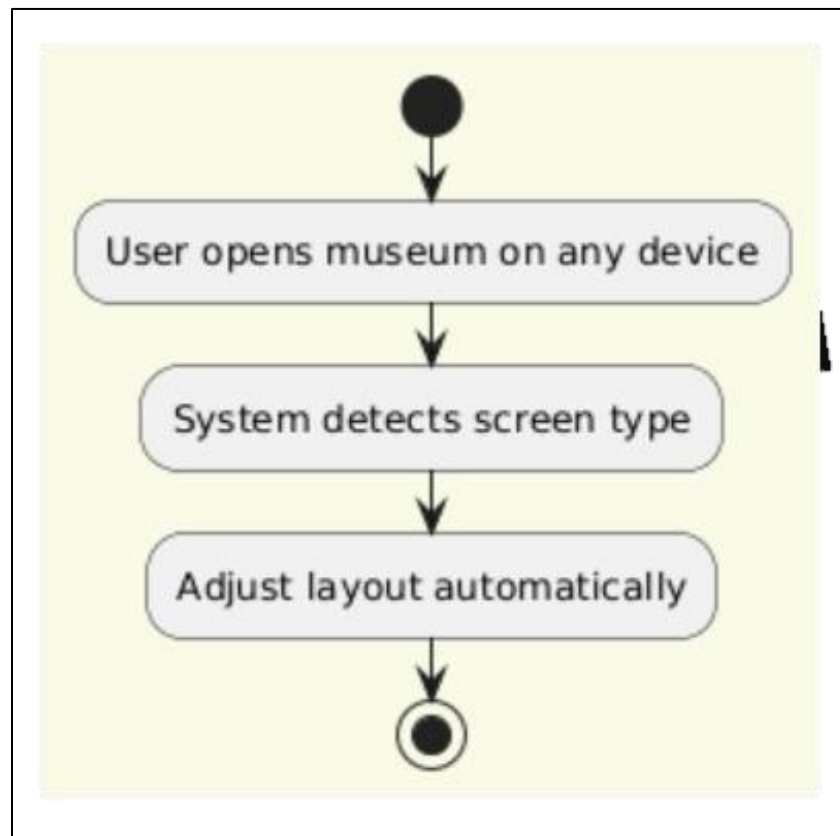


Figure 20: Responsive

7.4 Sequence Diagram

7.4.1 Registration

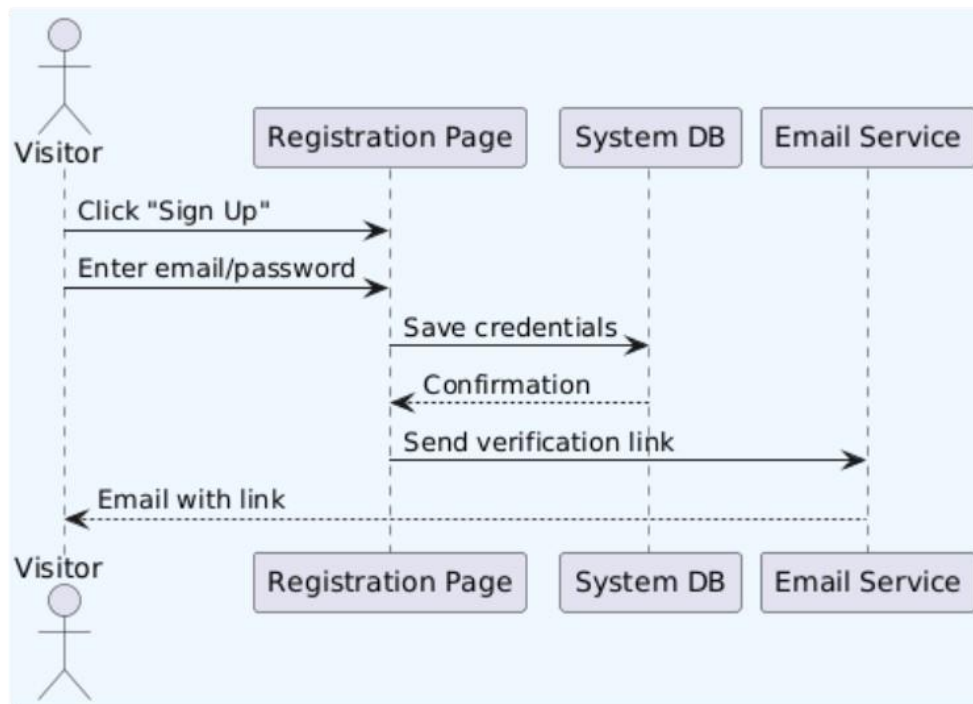


Figure 21: Registration

7.4.2 Email Verification

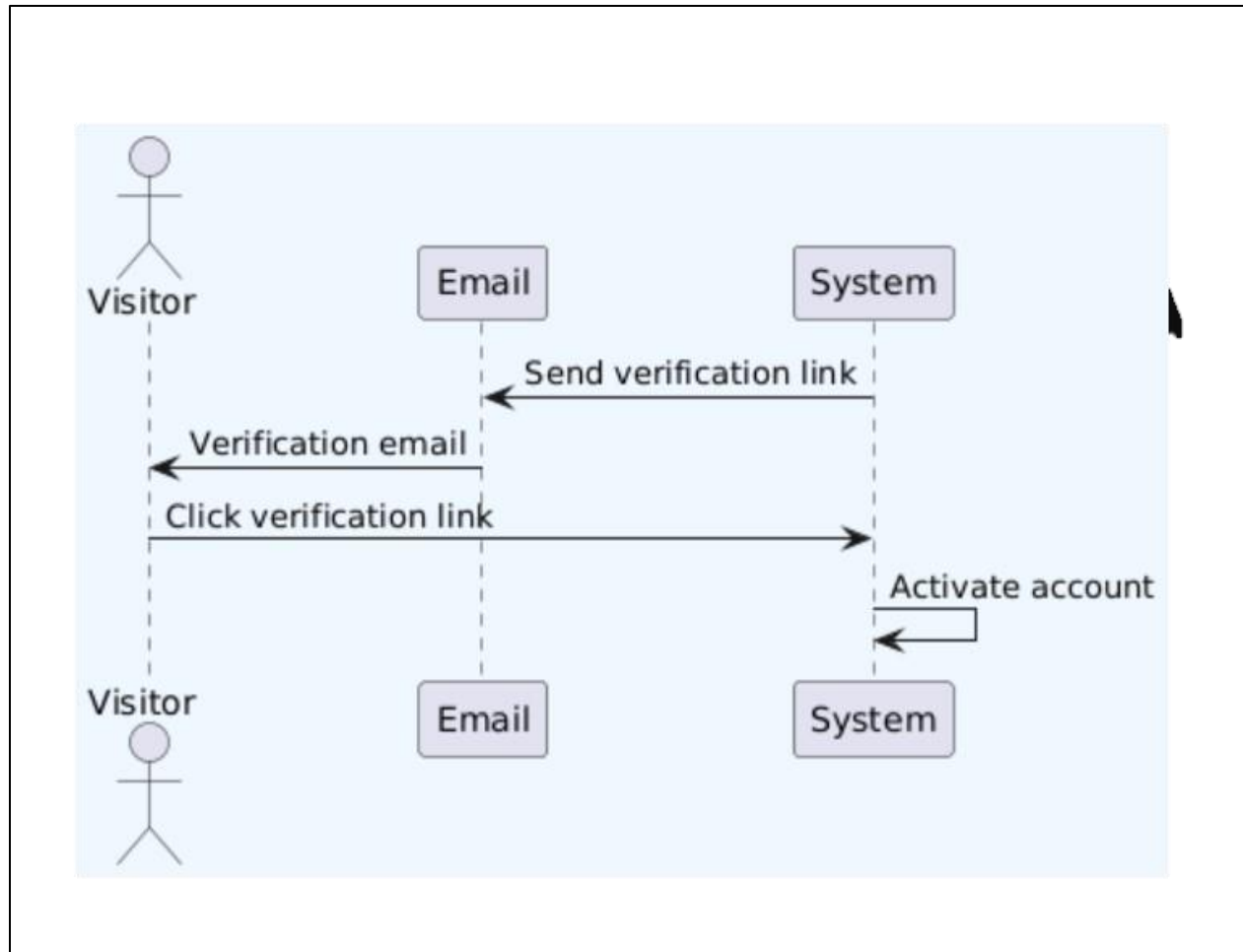


Figure 22: Email Verification

7.4.3 Login

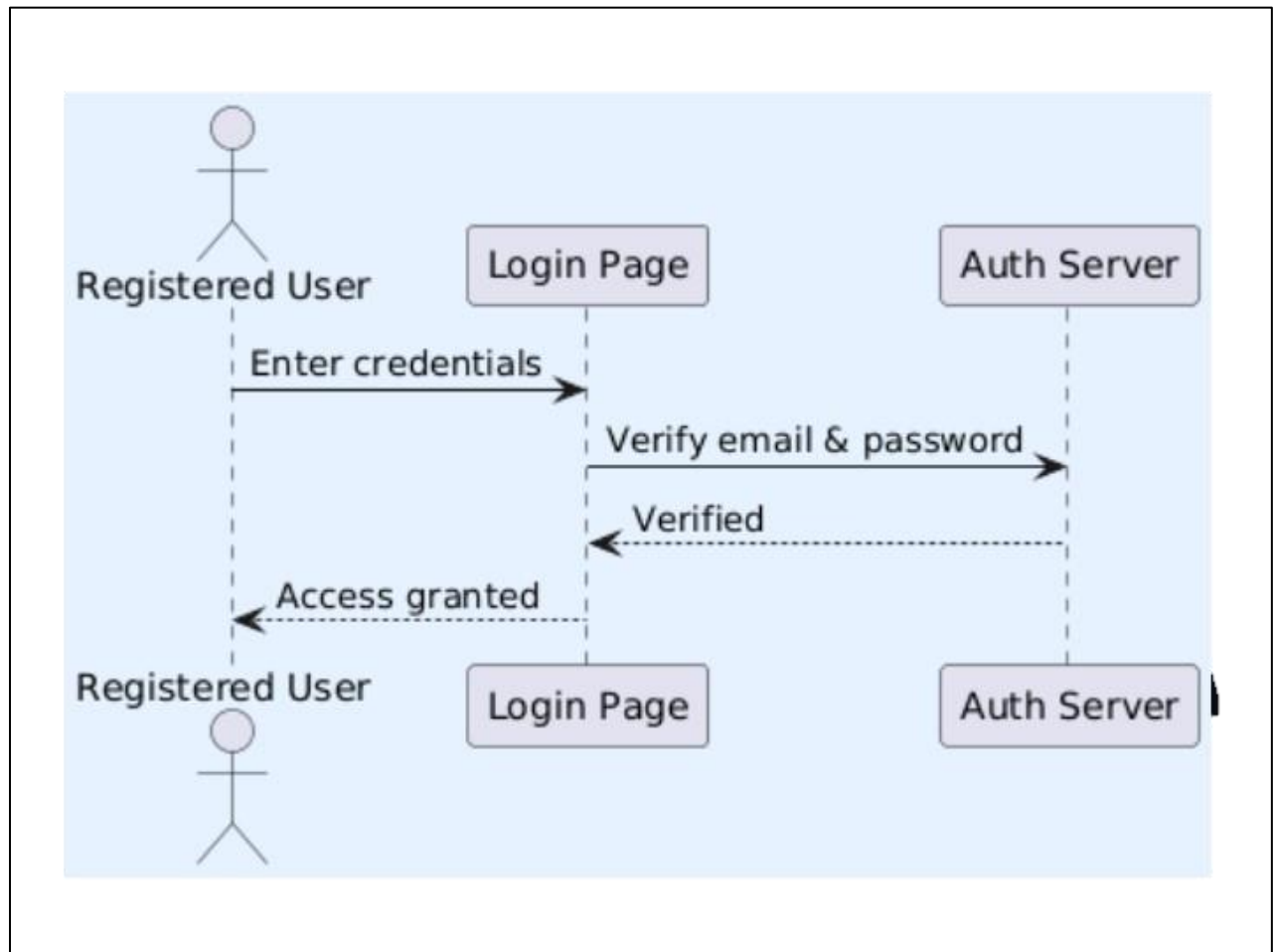


Figure 23: Login

7.4.4 Reset Password

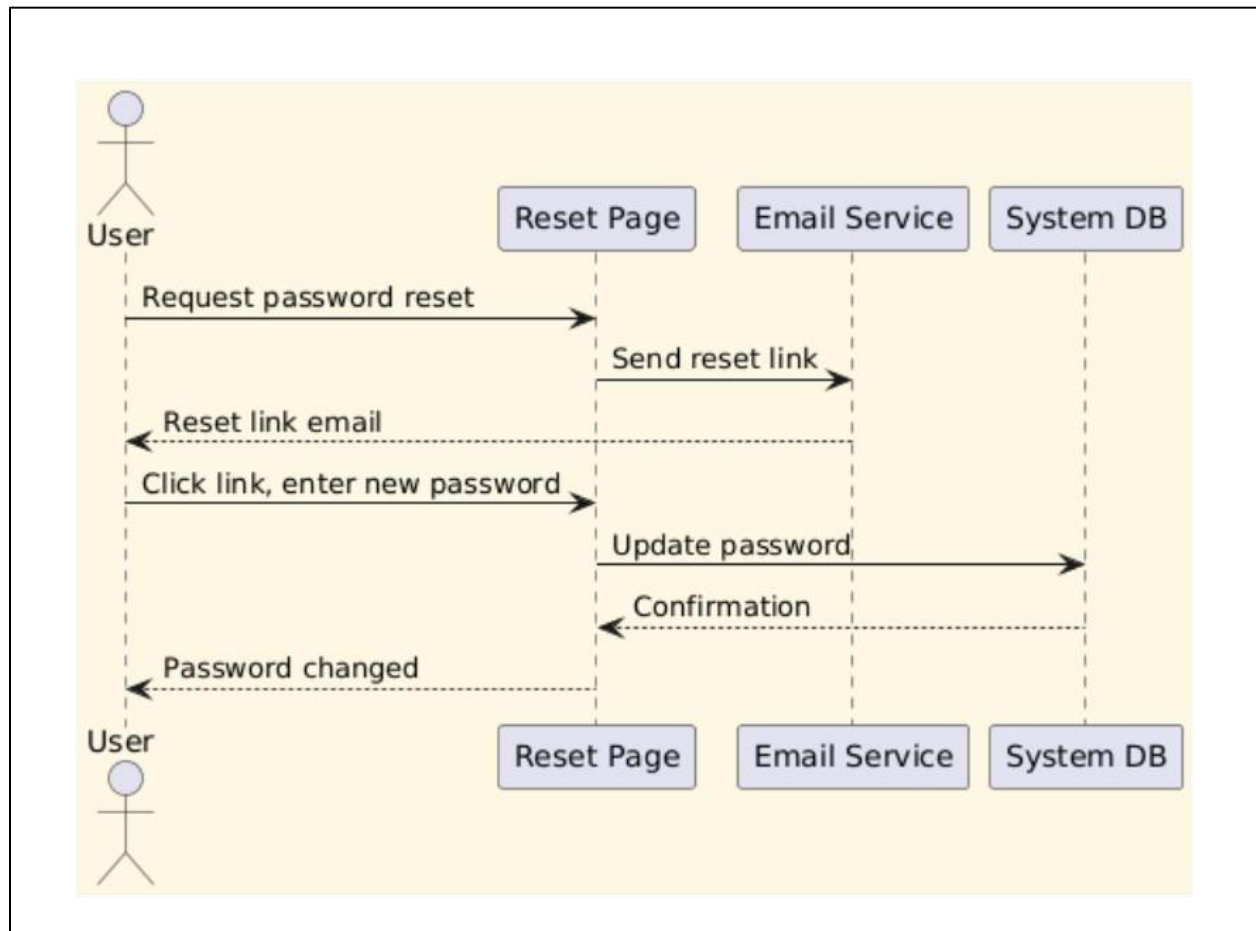


Figure 24: Reset Password

7.4.5 Navigate Museum

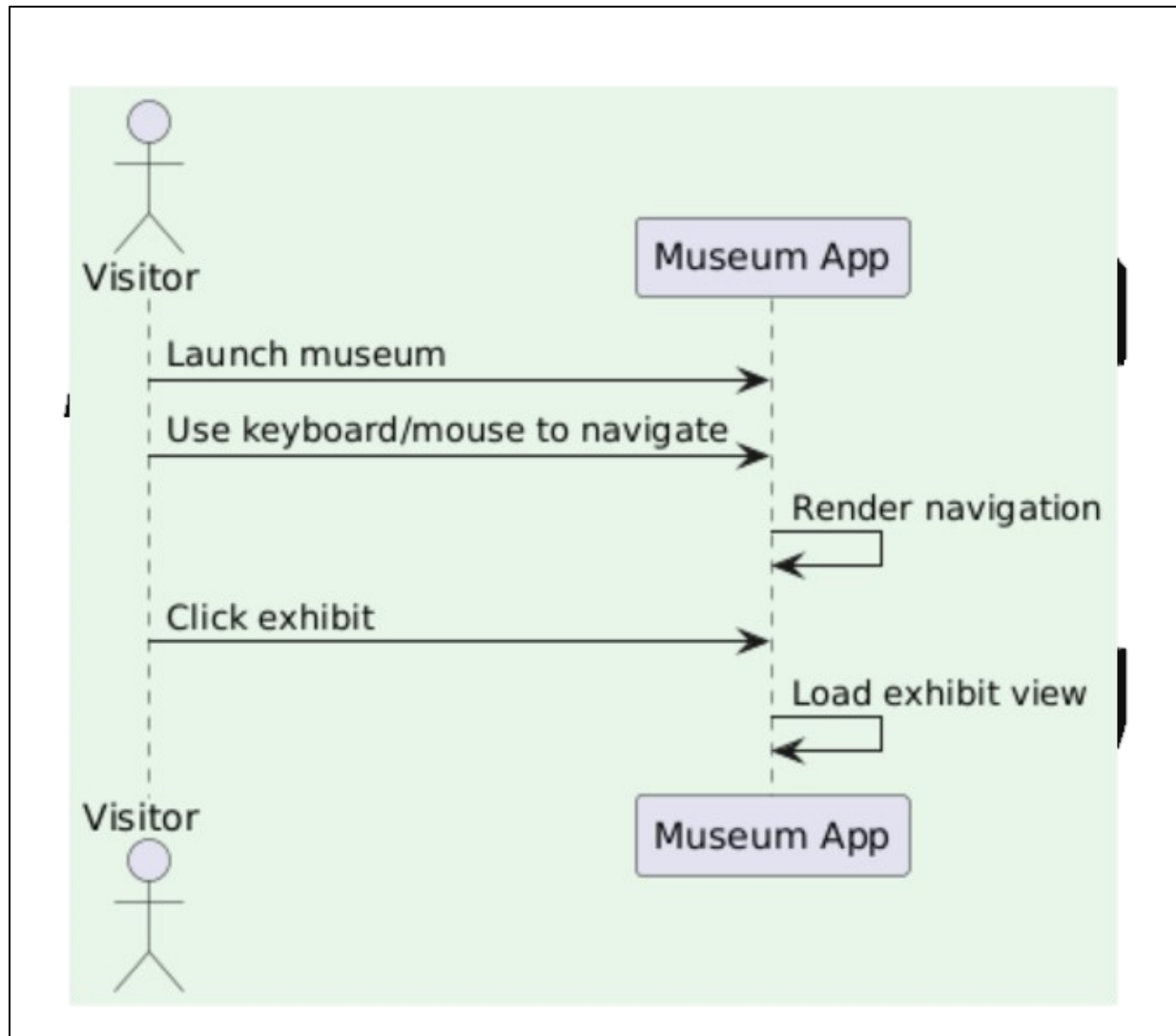


Figure 25: Navigate Museum

7.4.6 View Museum

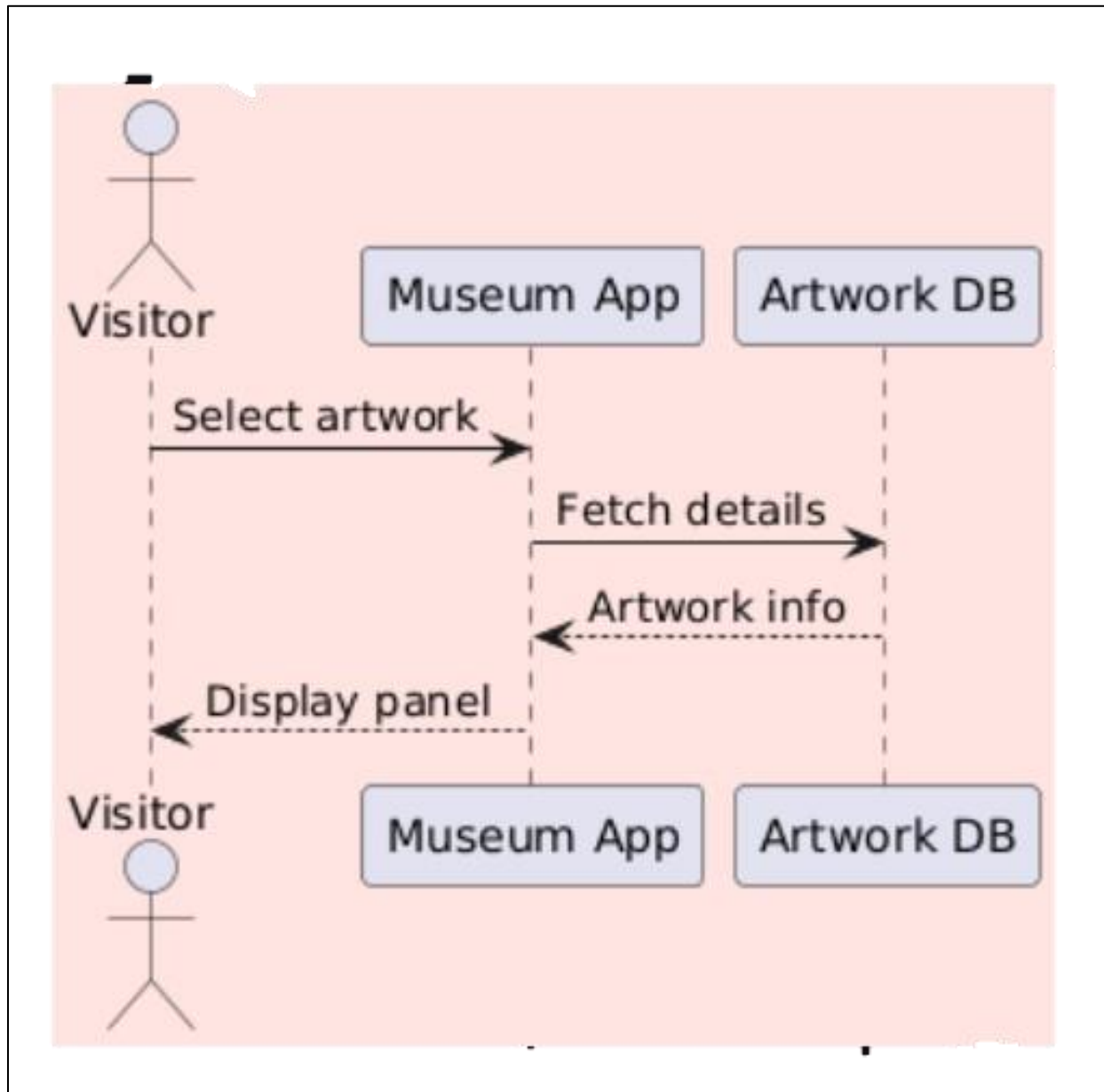


Figure 26: View Museum

7.4.7 View 3D

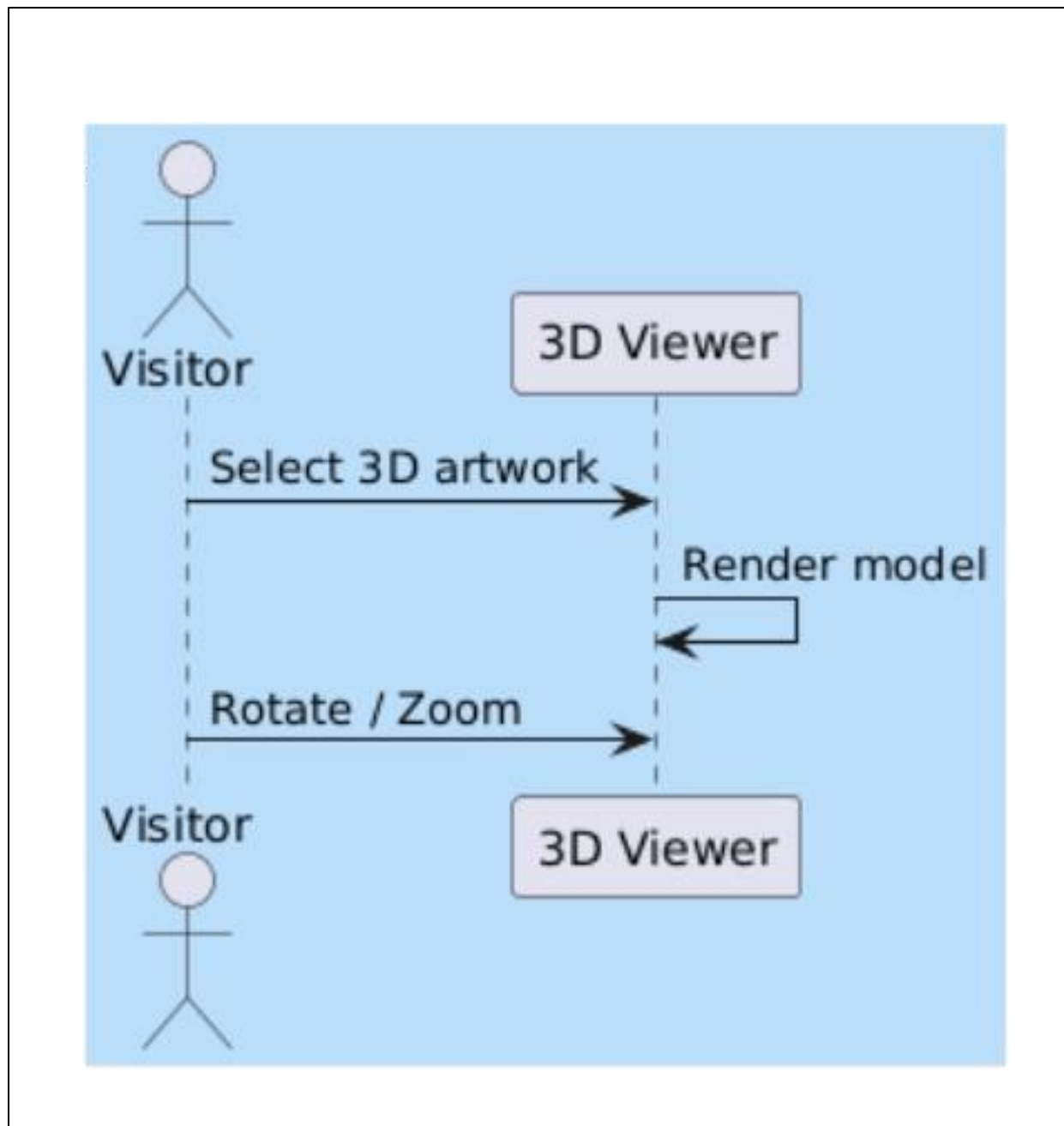


Figure 27: View 3D

7.4.8 Search

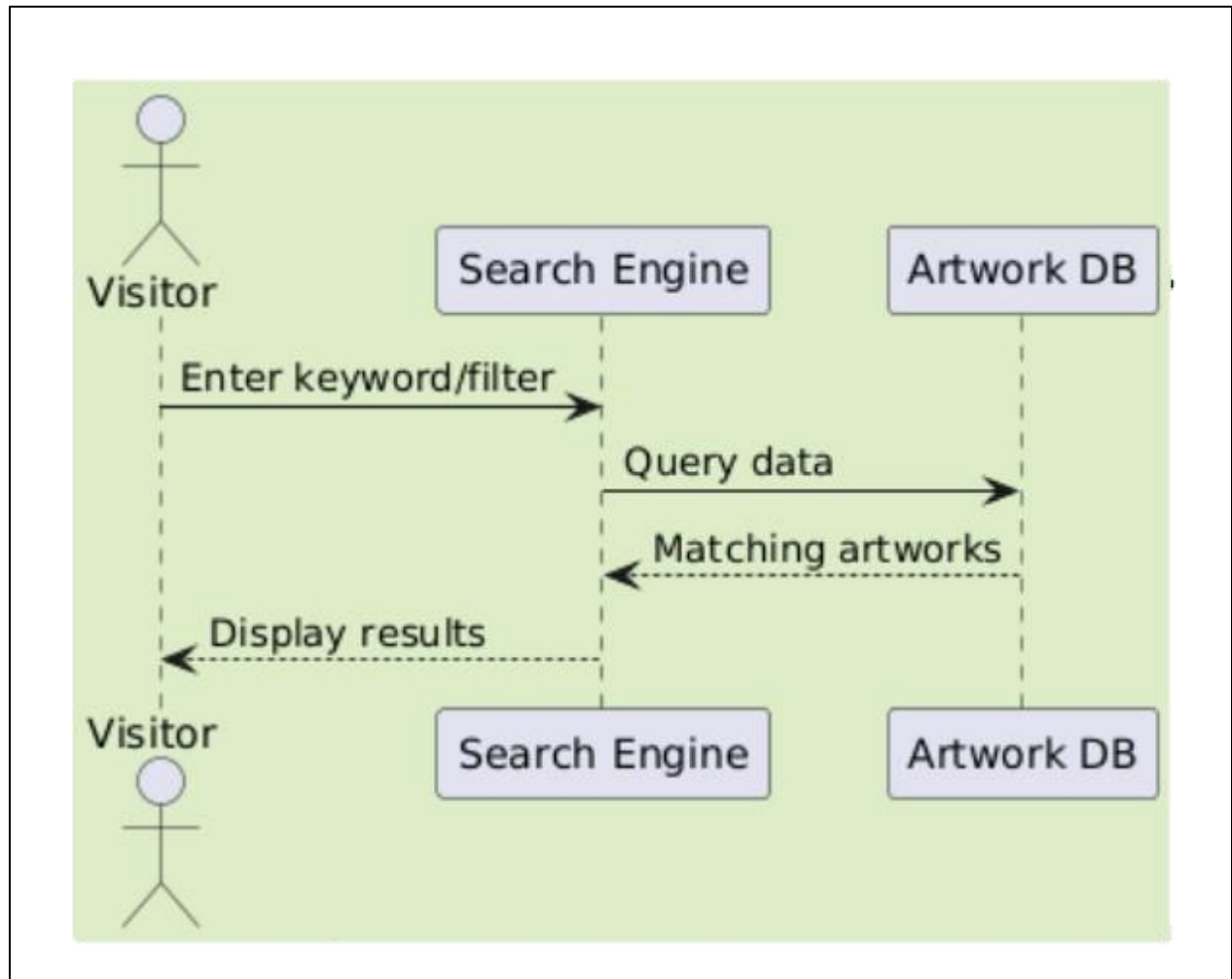


Figure 28: Search

7.4.9 Change Language

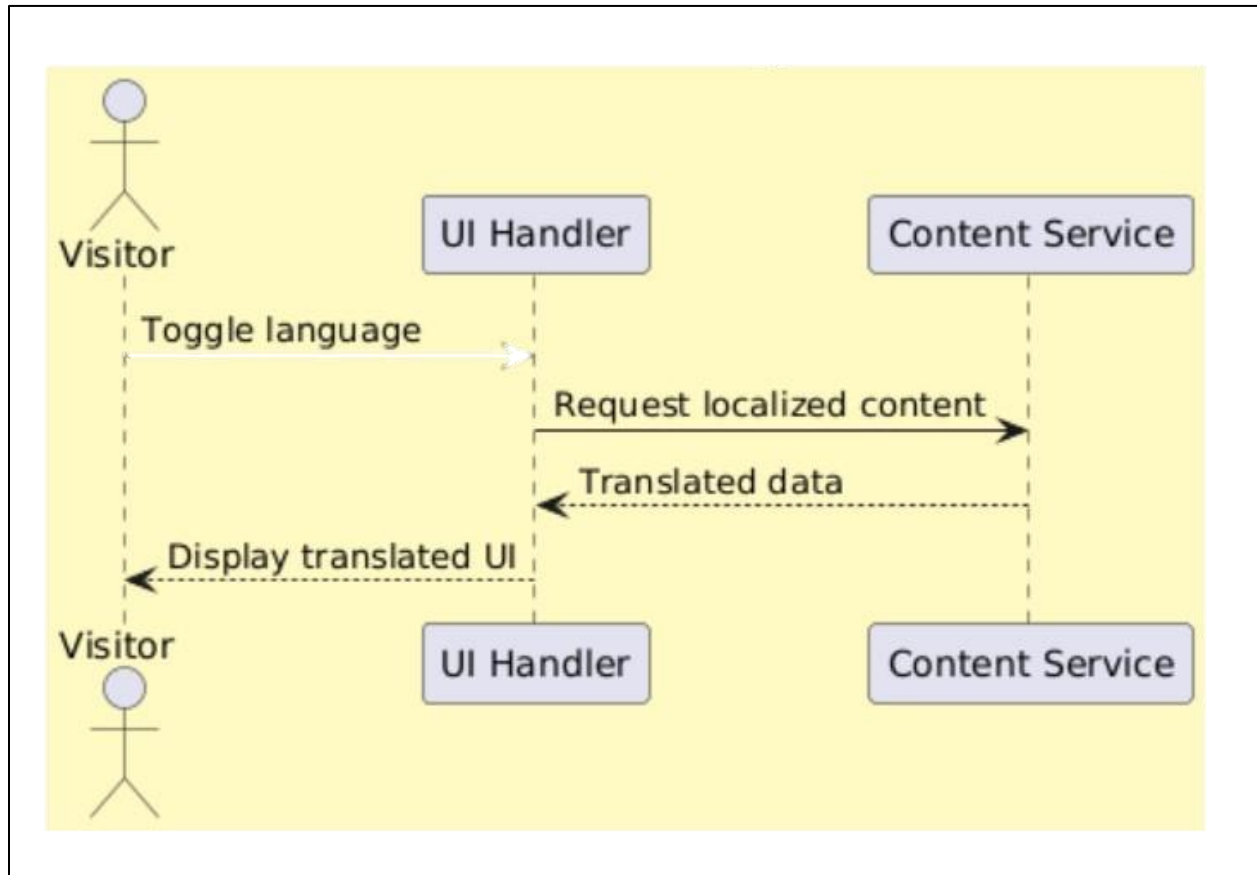


Figure 29: Change Language

7.4.10 Bookmark

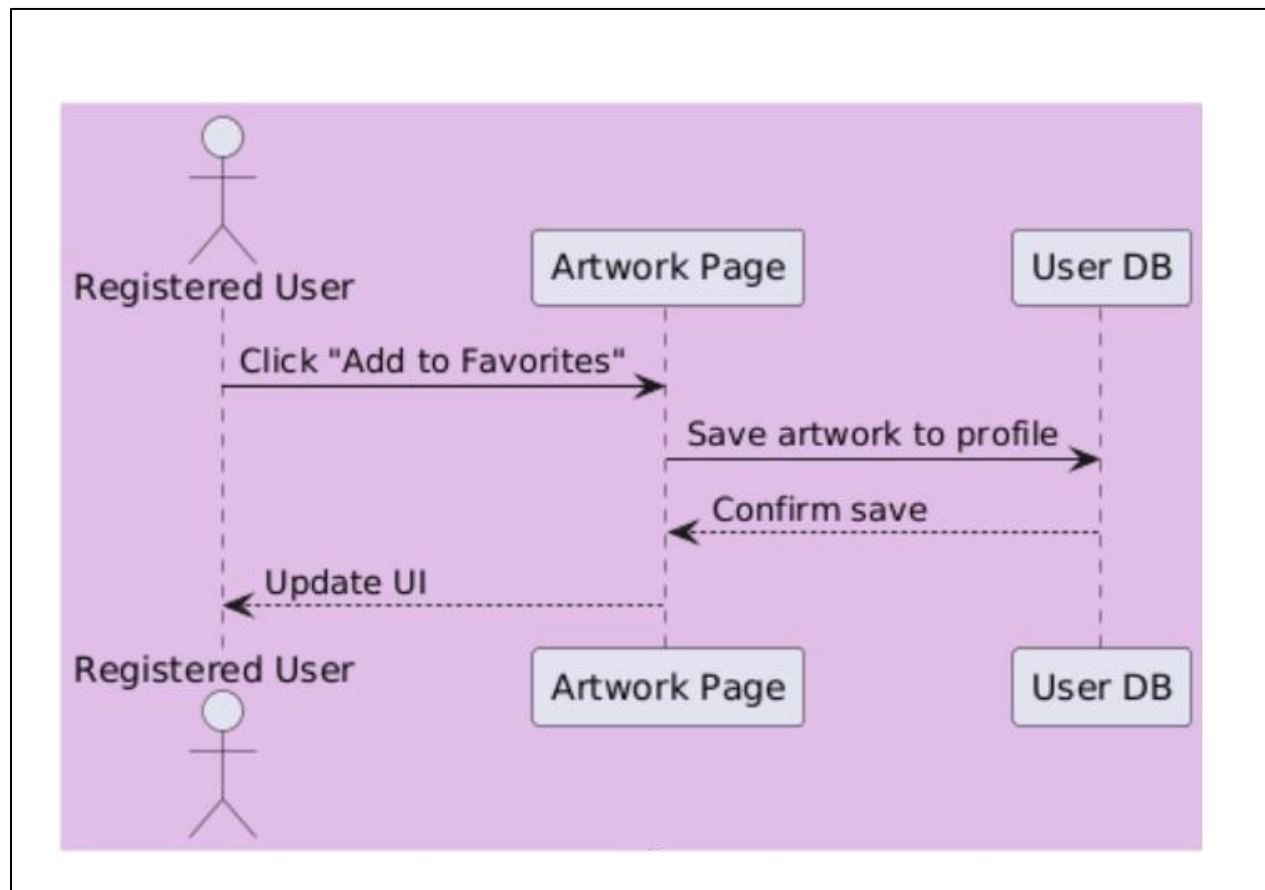


Figure 30: Bookmark

7.4.11 Audio Direction

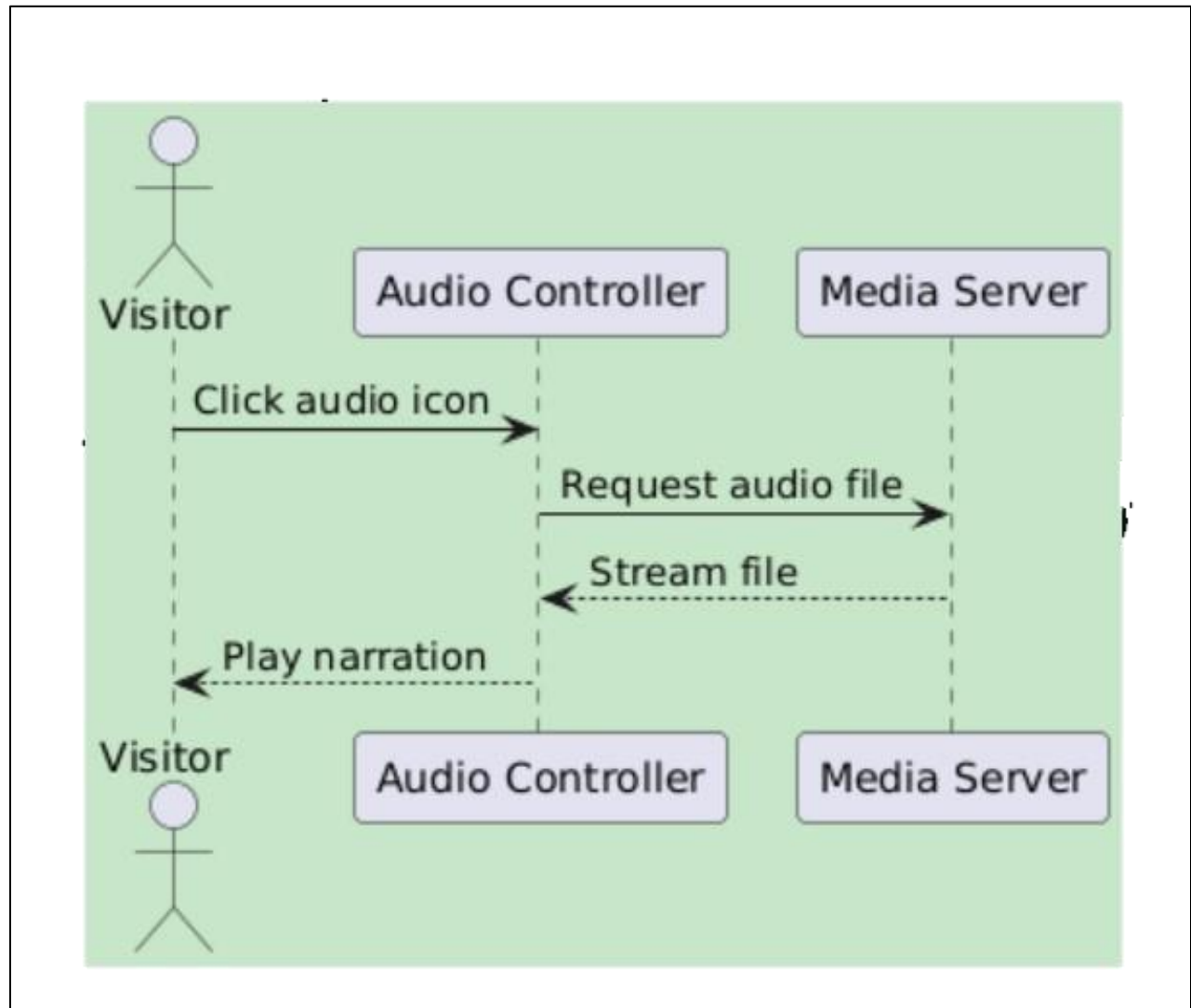


Figure 31: Audio Direction

7.4.12 Admin Login

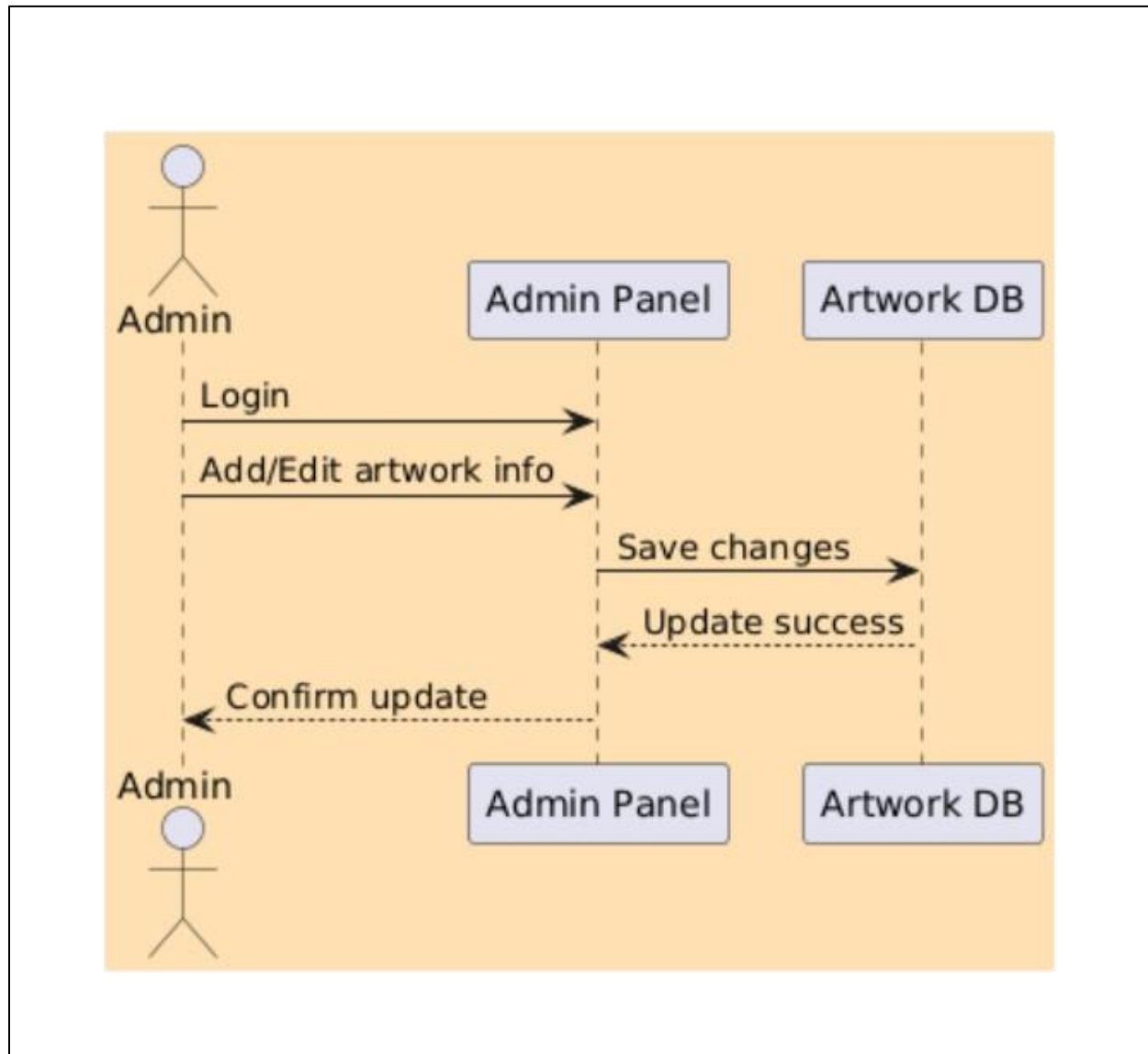


Figure 32: Admin Login

7.4.13 Reset Password

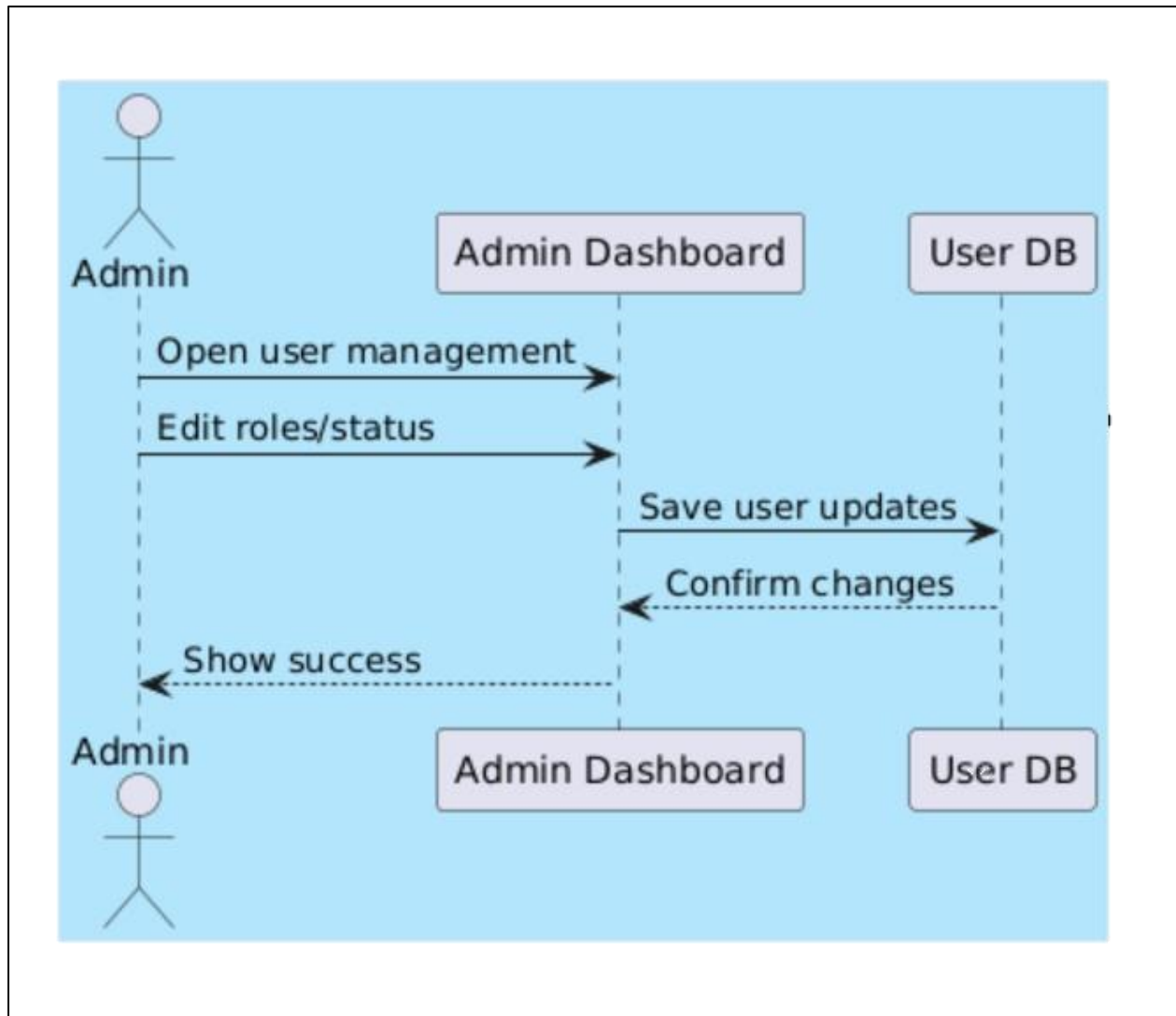


Figure 33: Admin Login

7.4.14 Submit Feedback

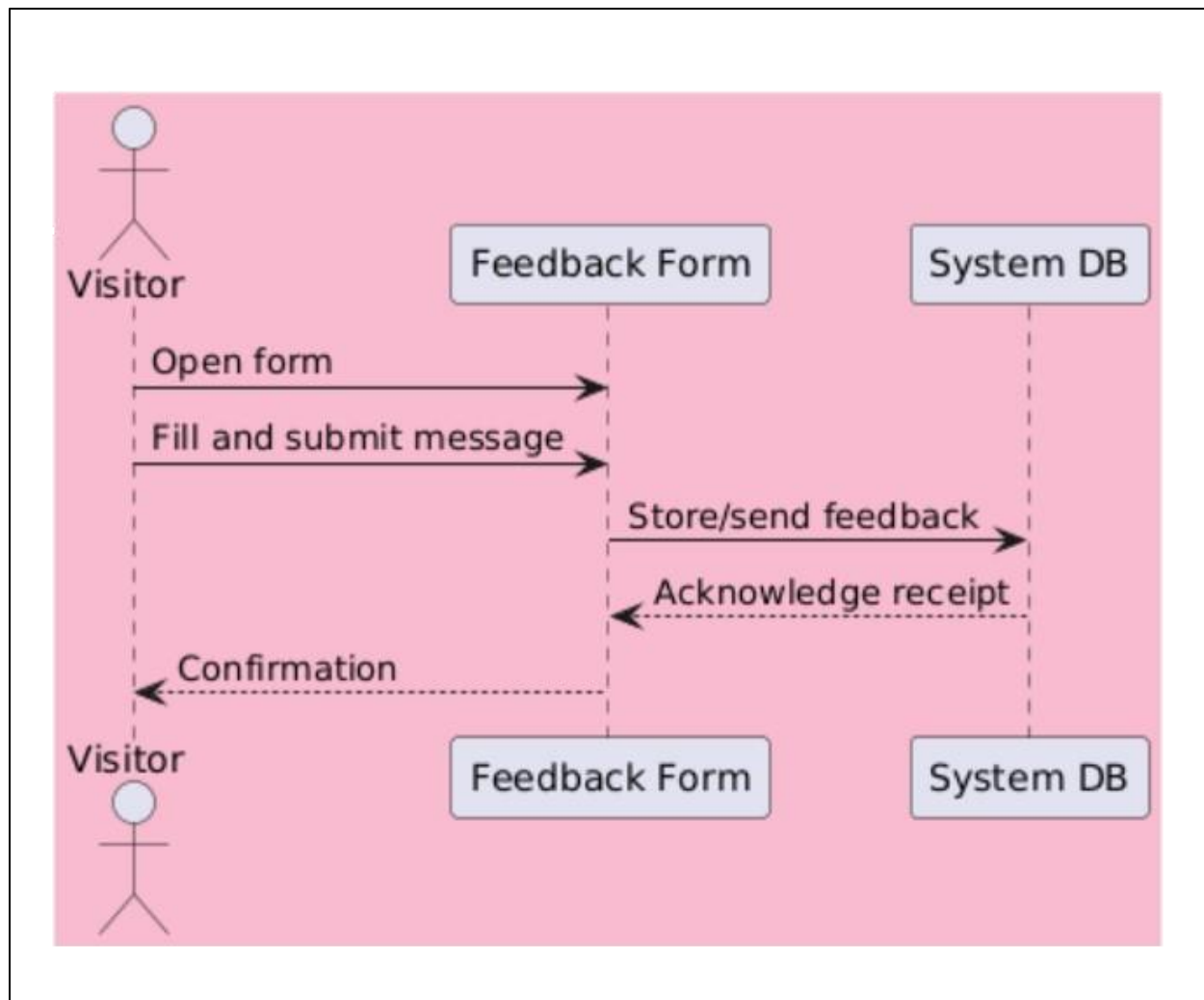


Figure 34: Submit Feedback

7.4.15 Visit Tracking

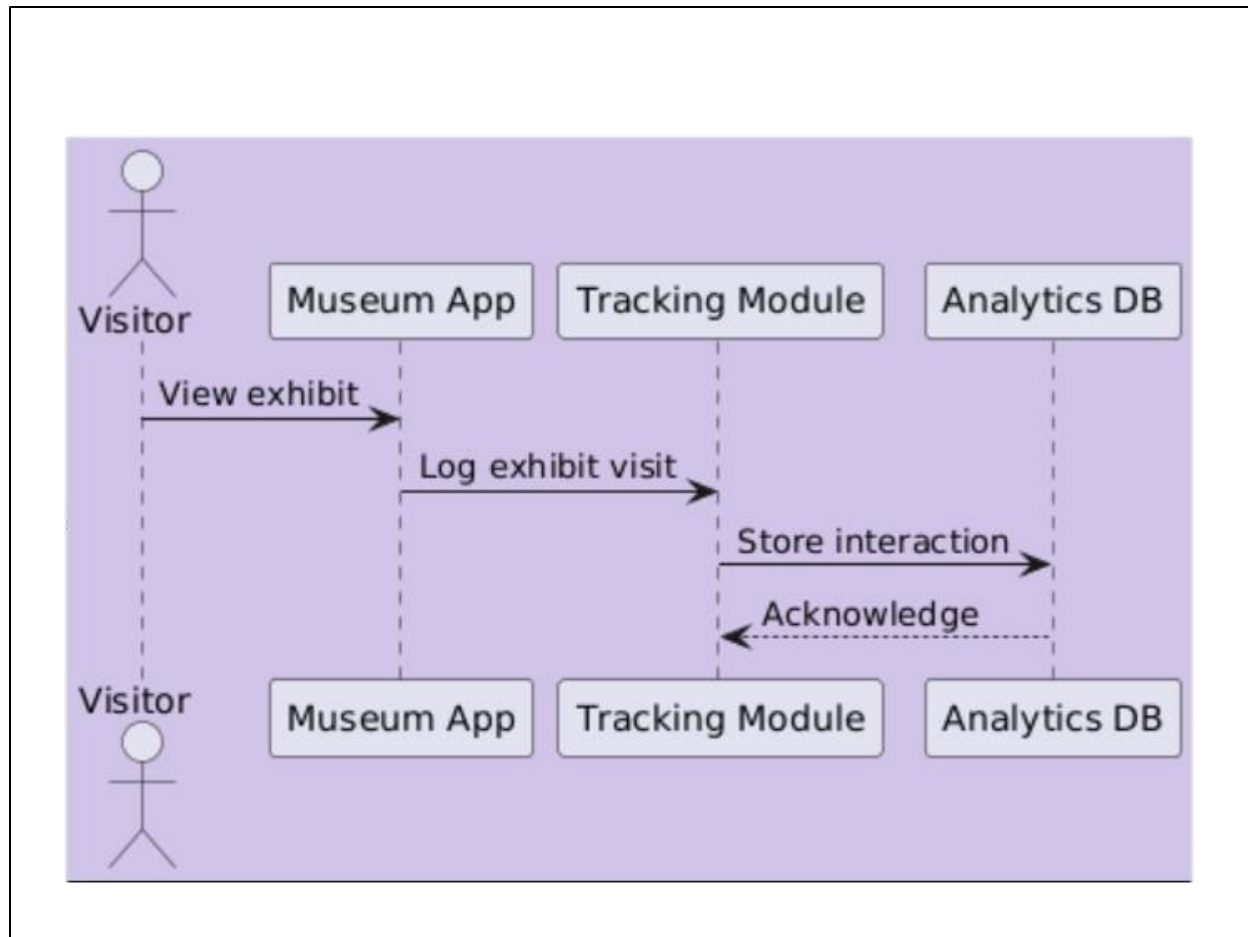


Figure 35: Visit Tracking

7.4.16 Artefacts Presentation

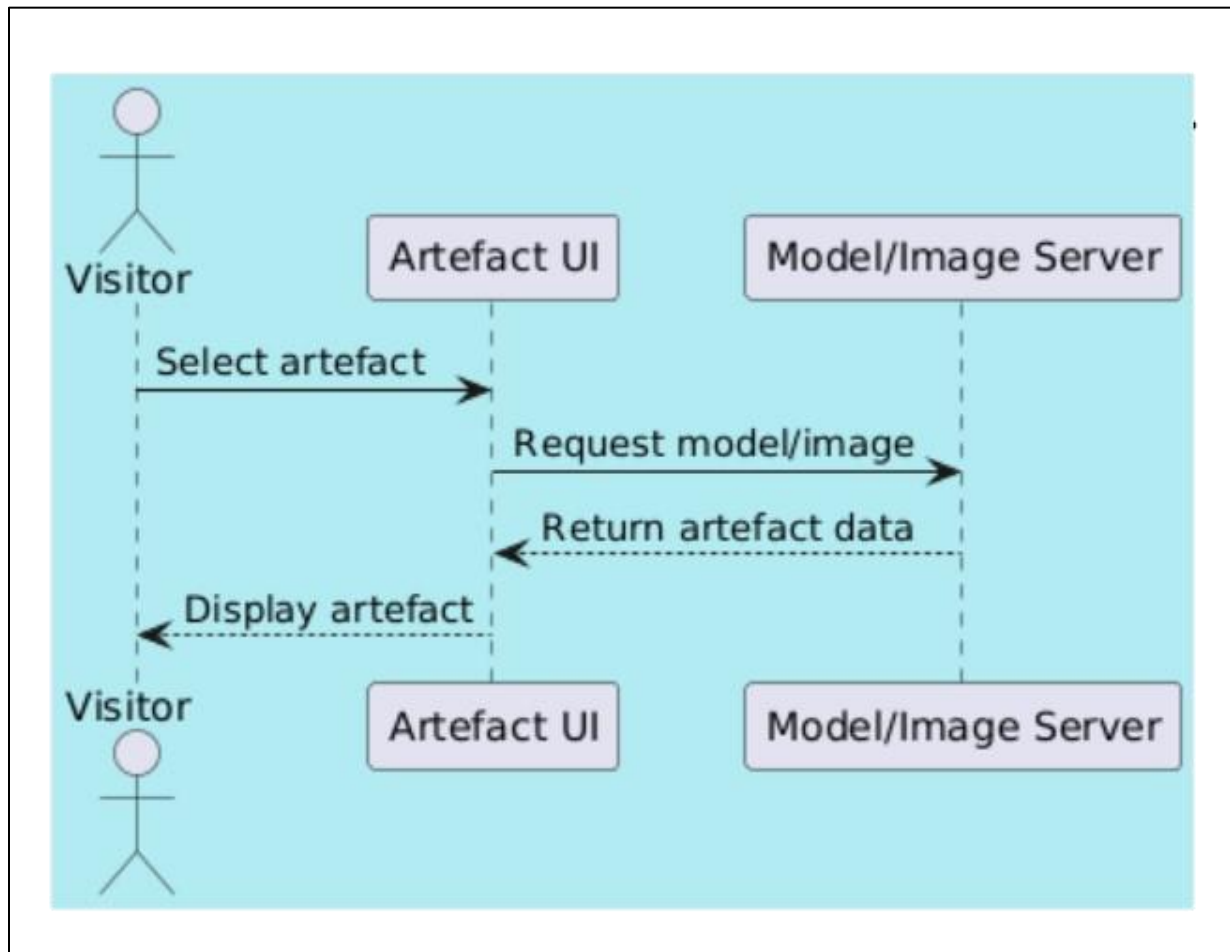


Figure 36: Artefacts Presentation

7.4.17 Print & Download

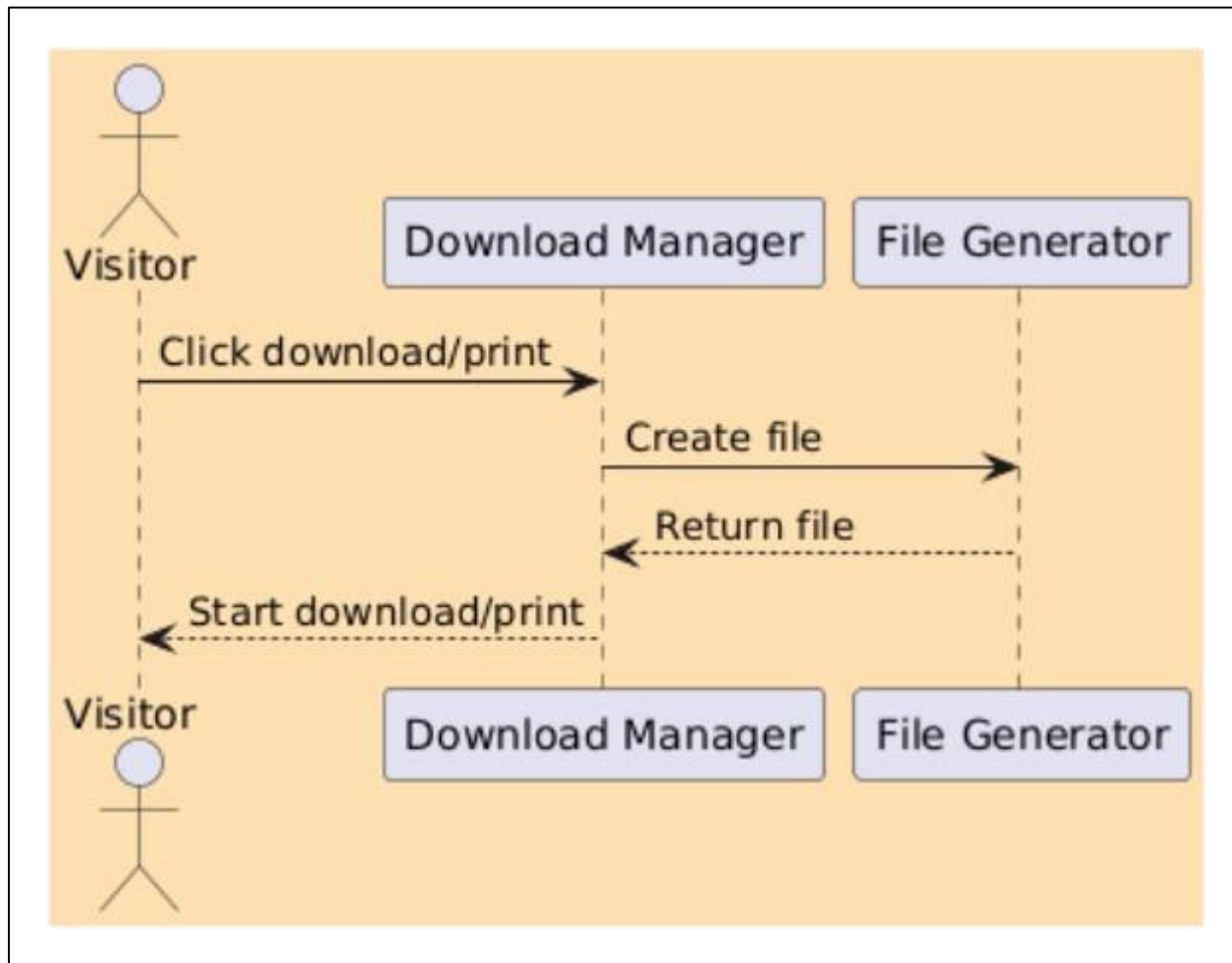


Figure 37: Print & Download

7.4.18 Responsive

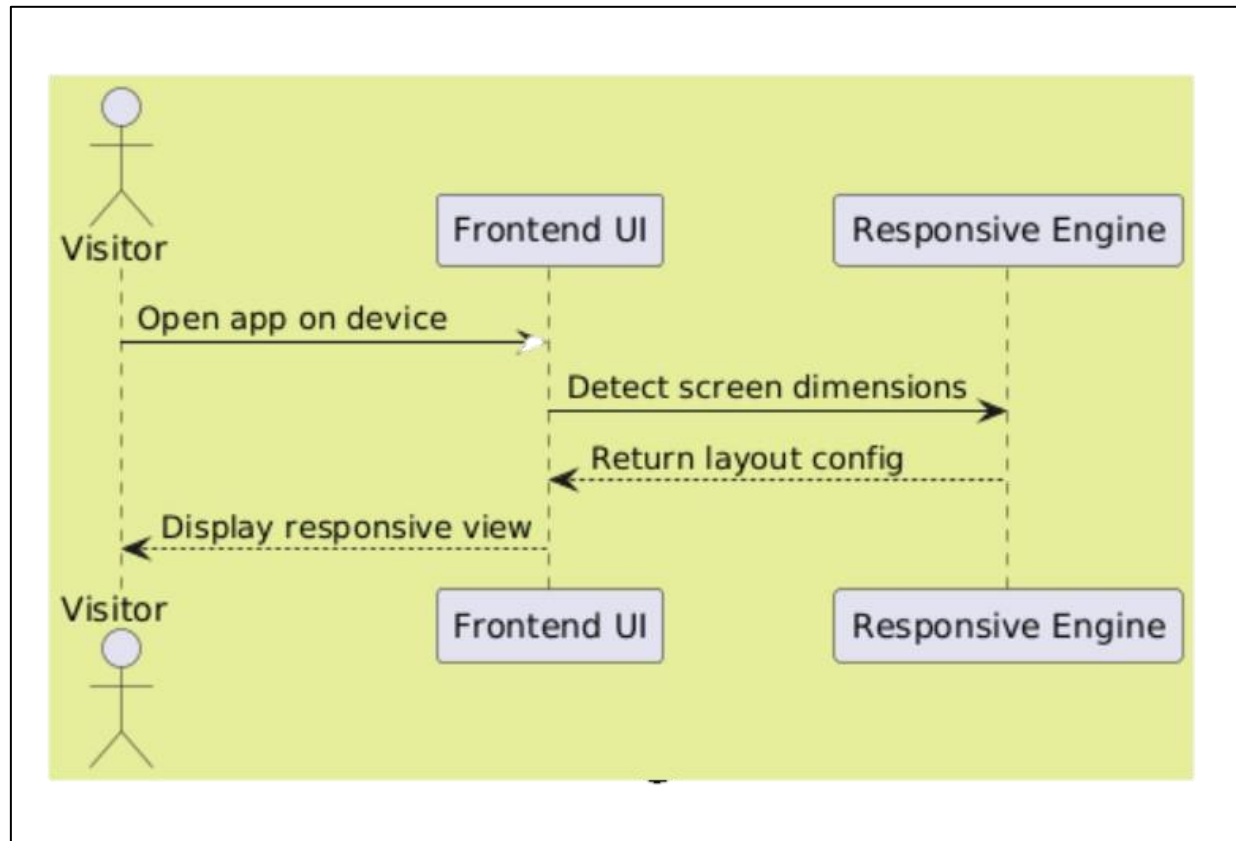


Figure 38: Responsive

7.5 Data Flow Diagram

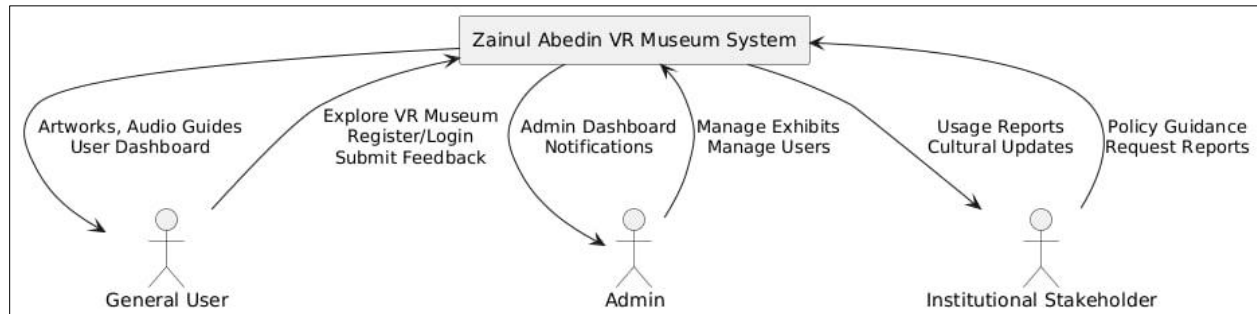


Figure 39: DFD Level 0

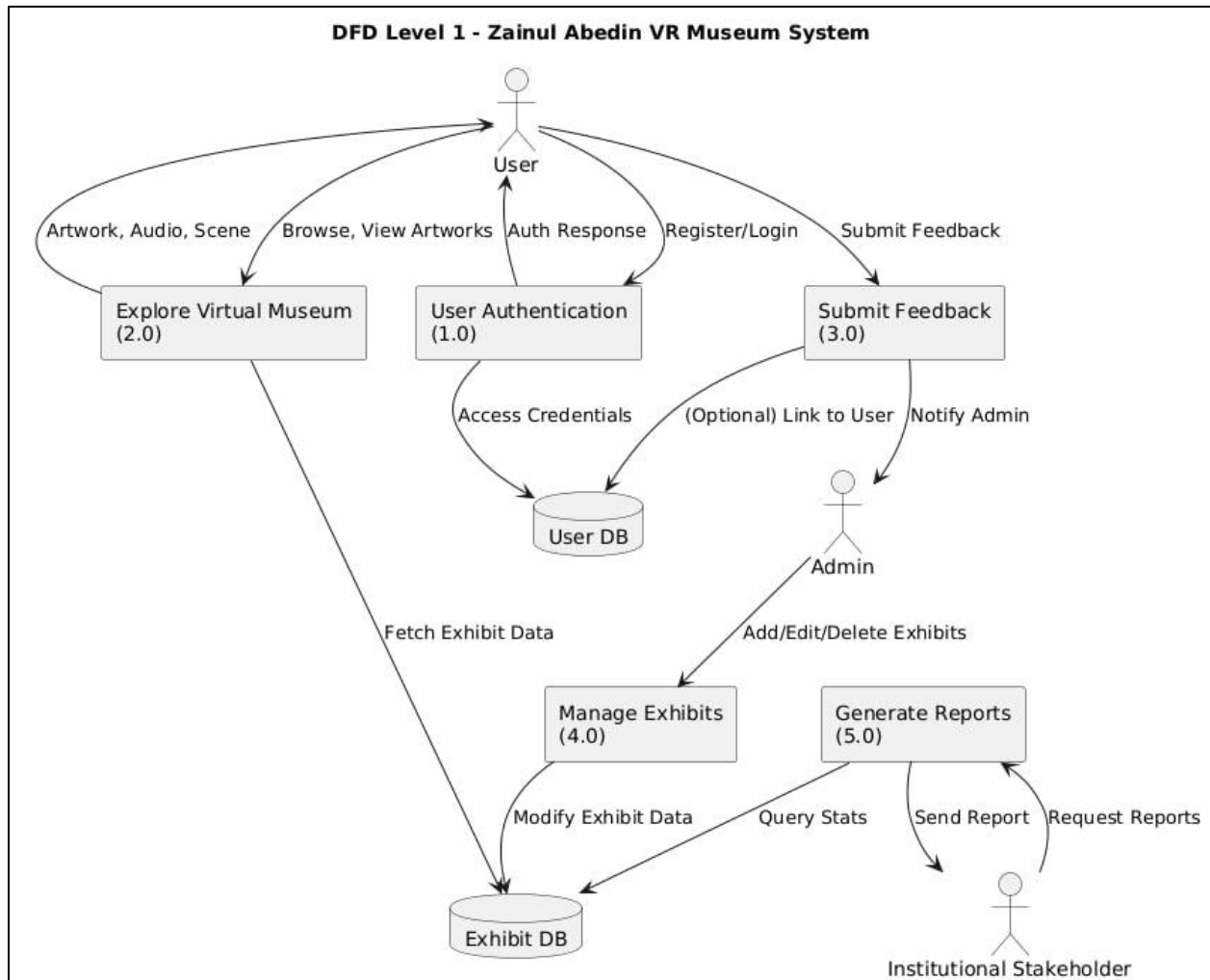


Figure 40: DFD Level 1

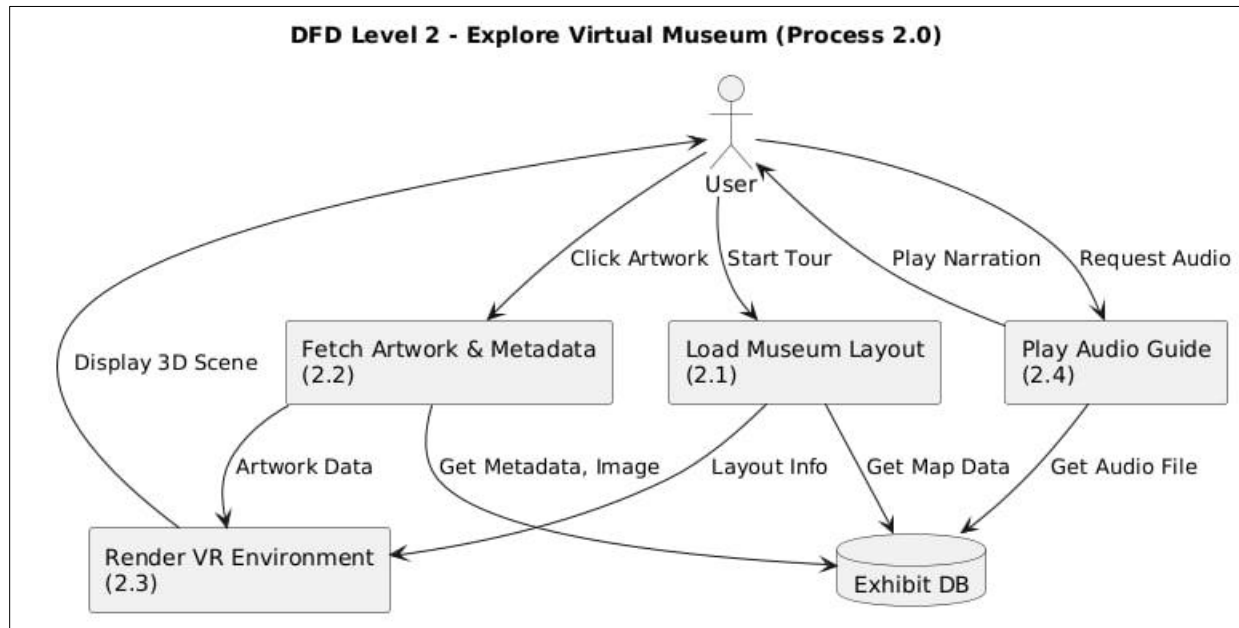
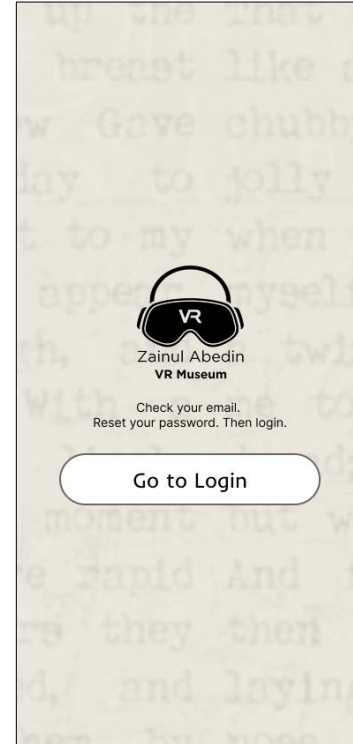
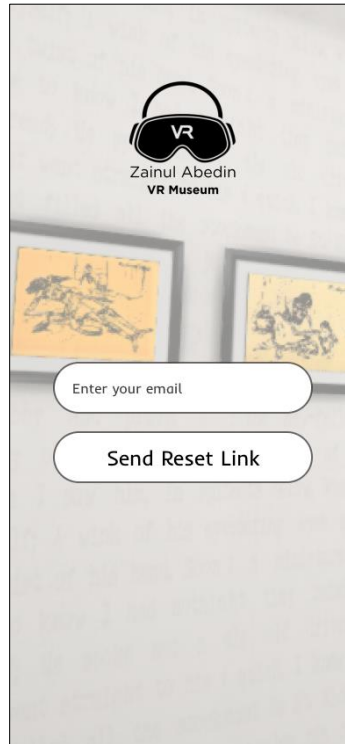
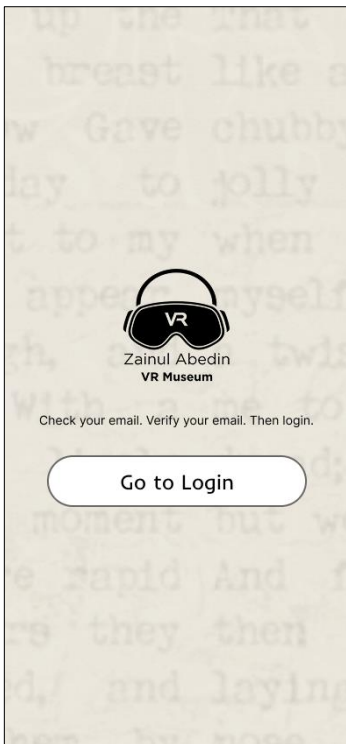
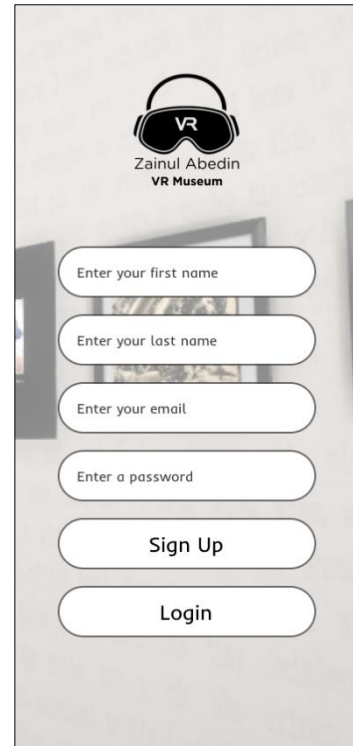
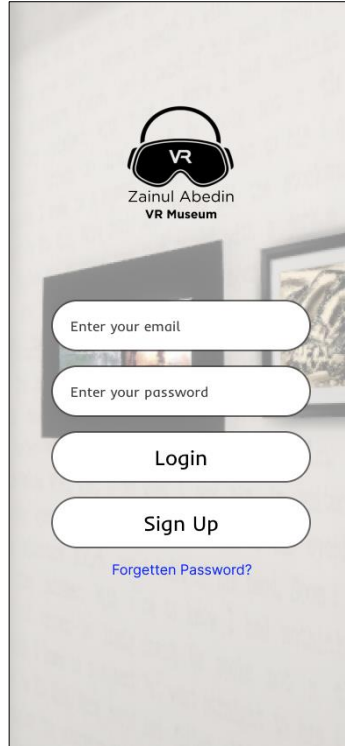
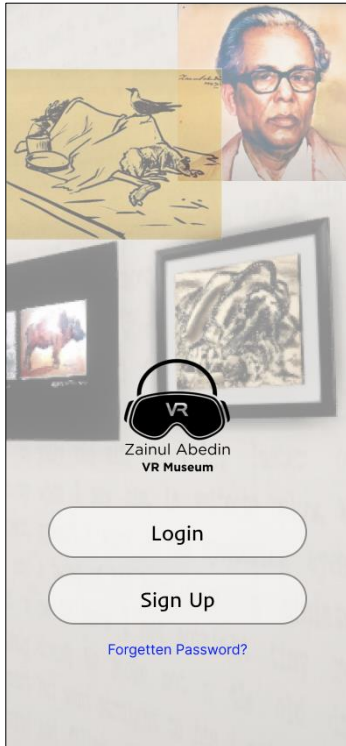
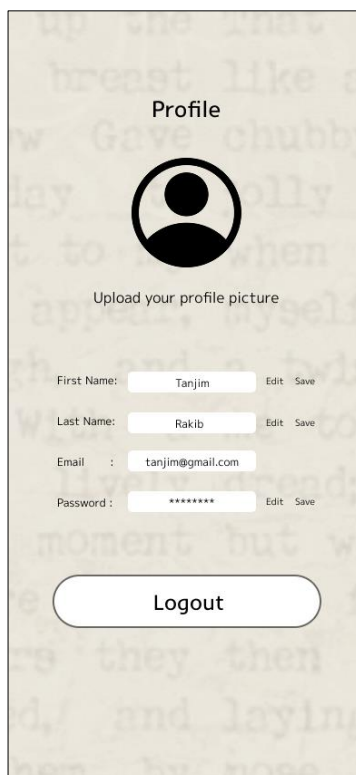
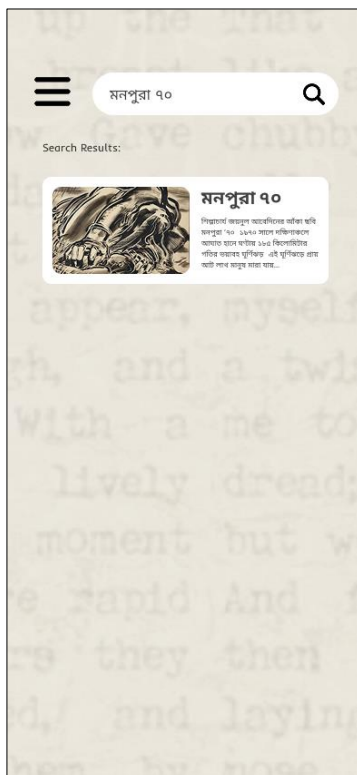
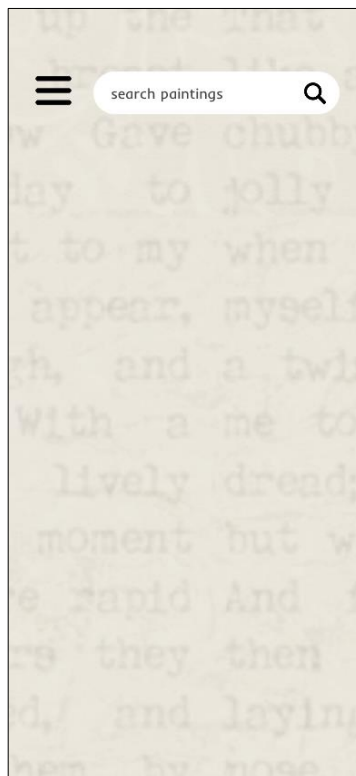


Figure 41: DFD Level 2

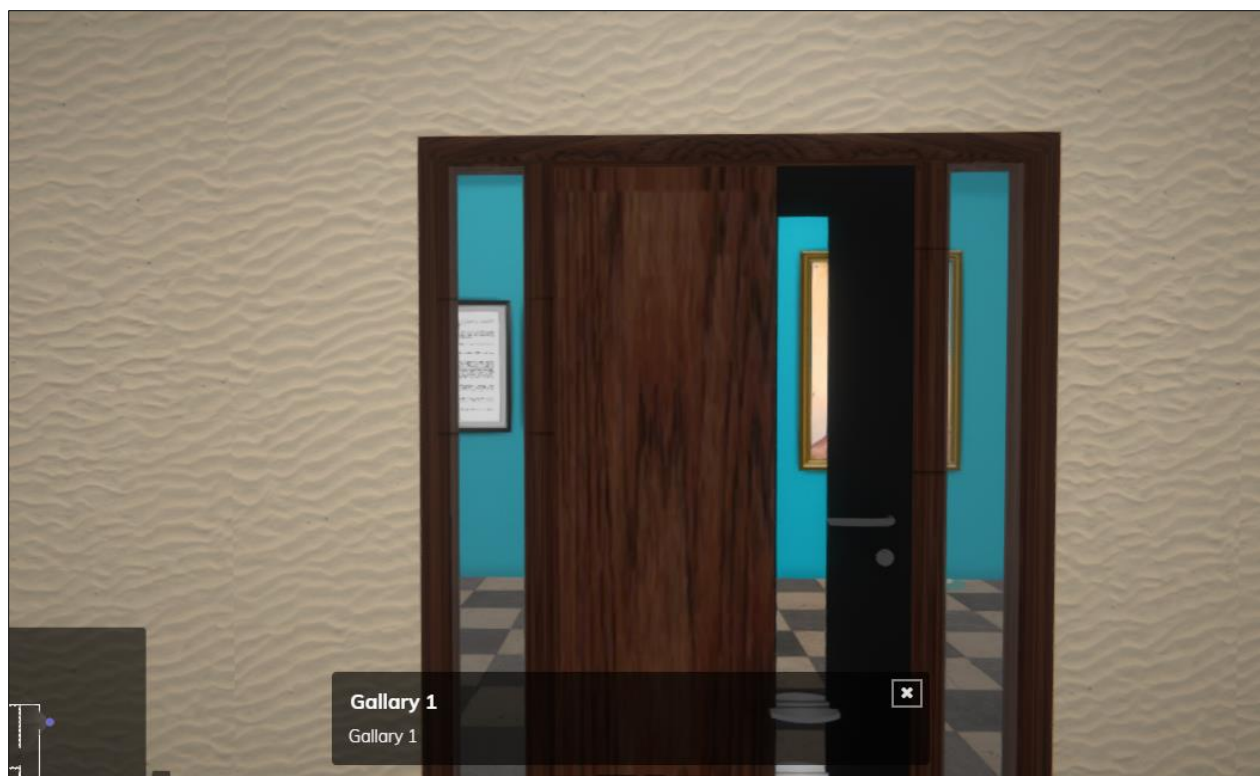
8. Prototyping

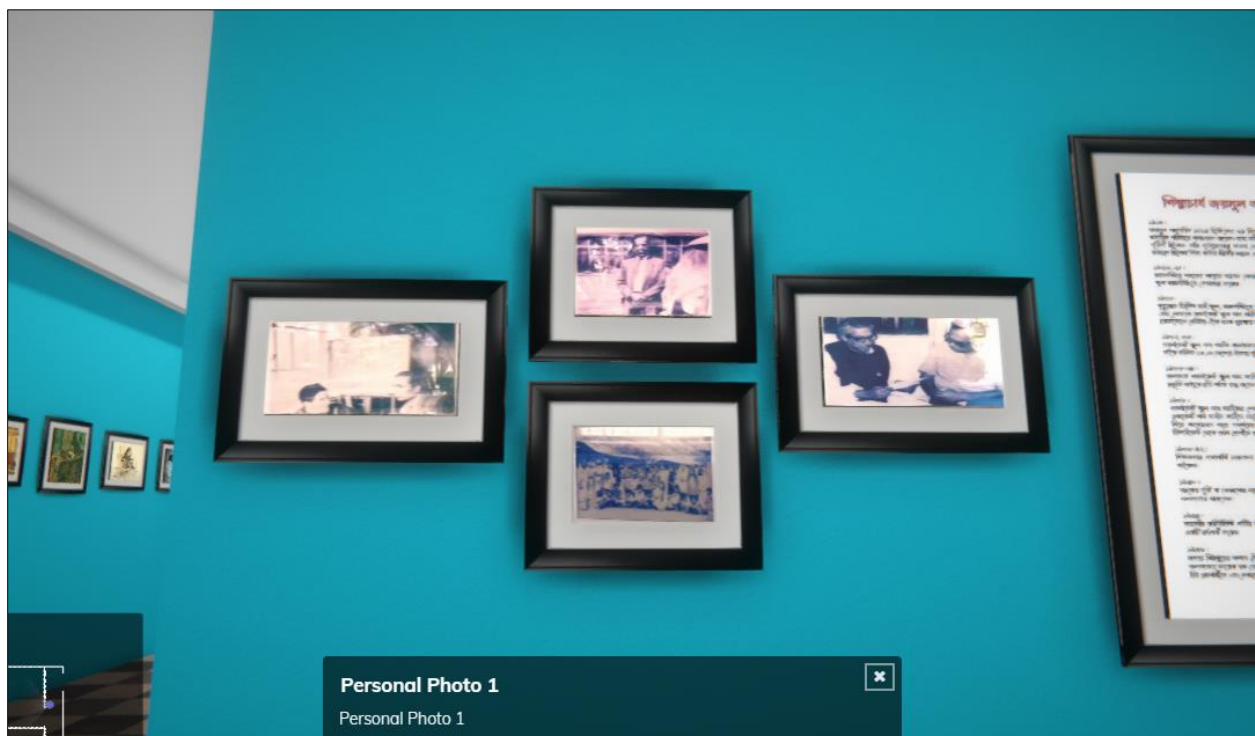
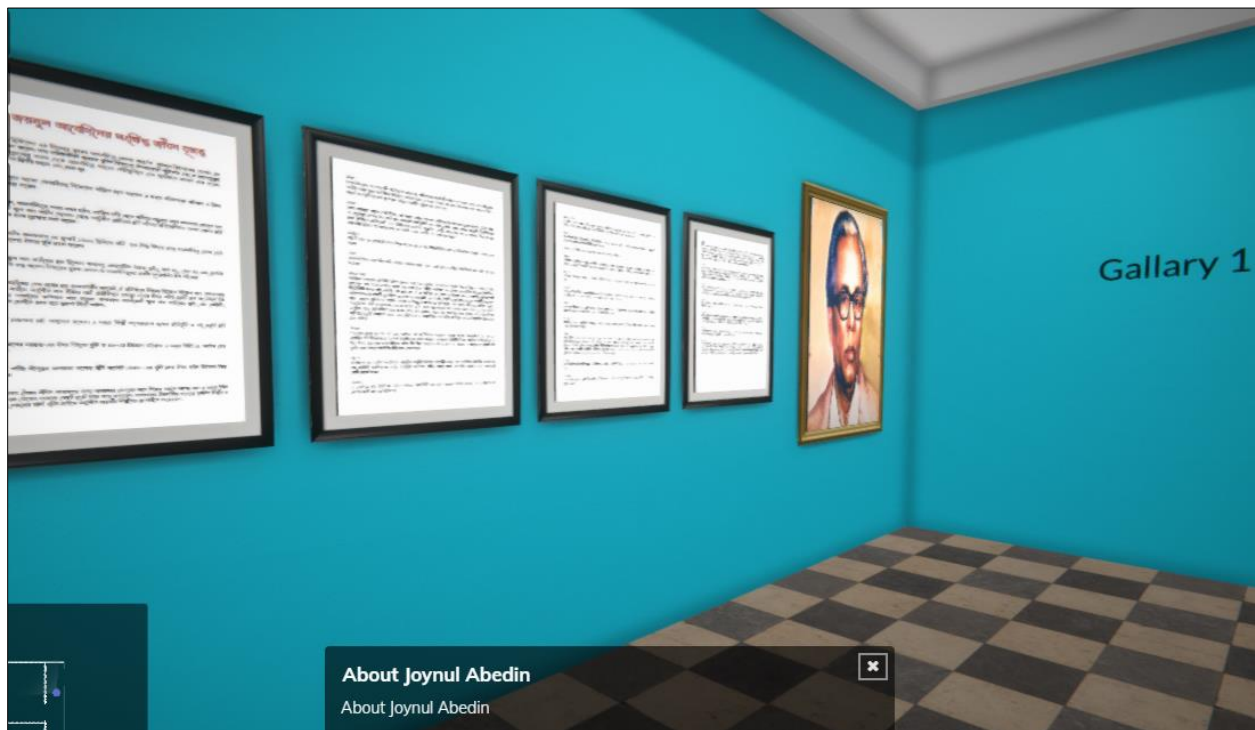
8.1 Low-Fidelity Prototype

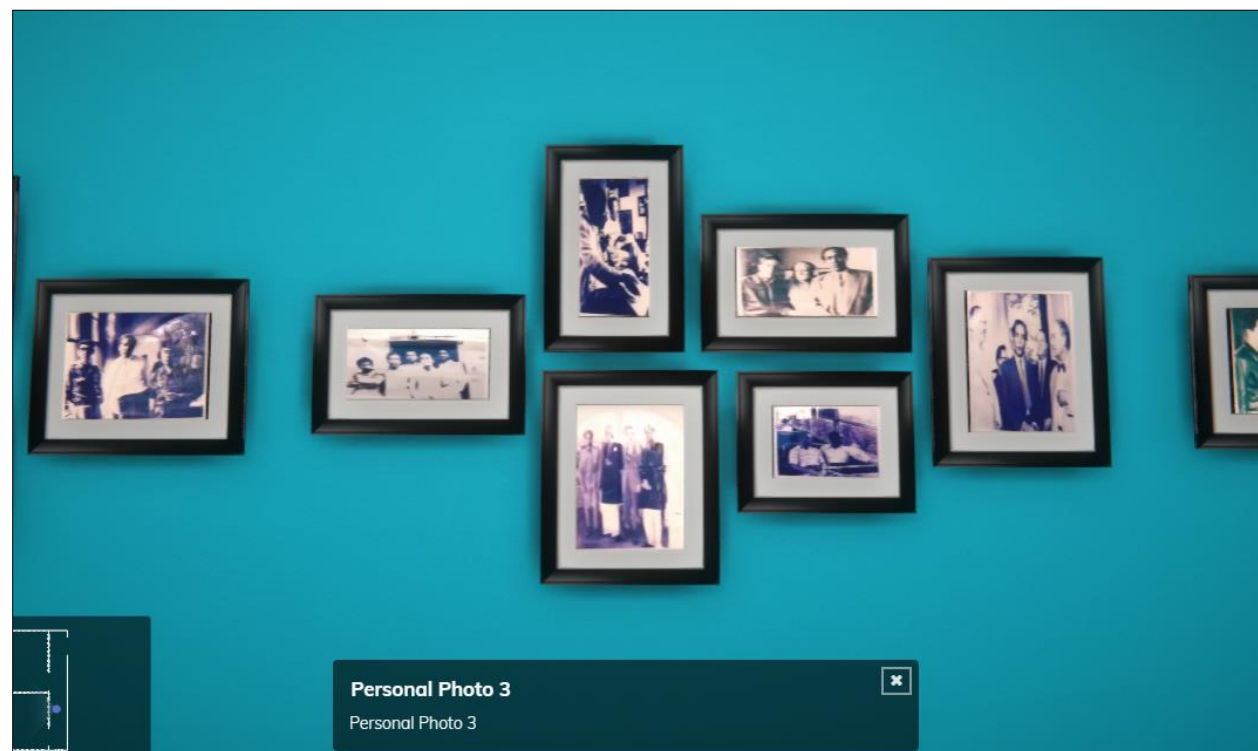
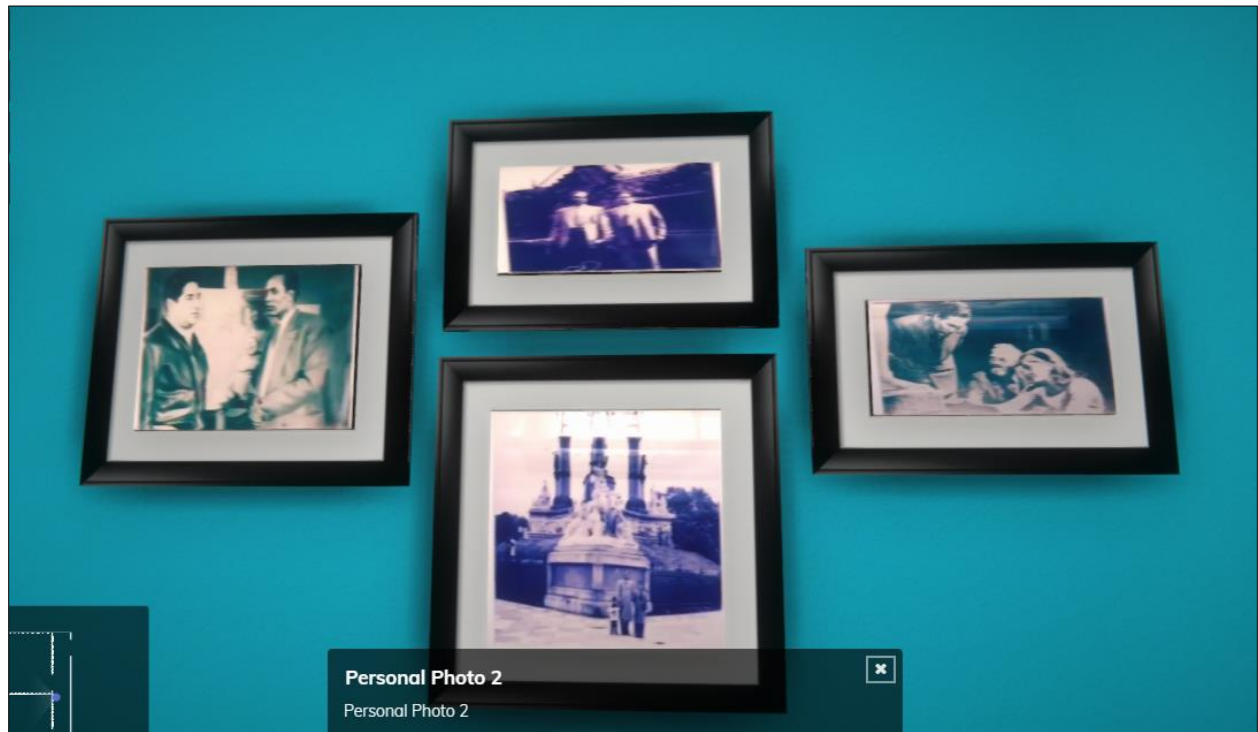


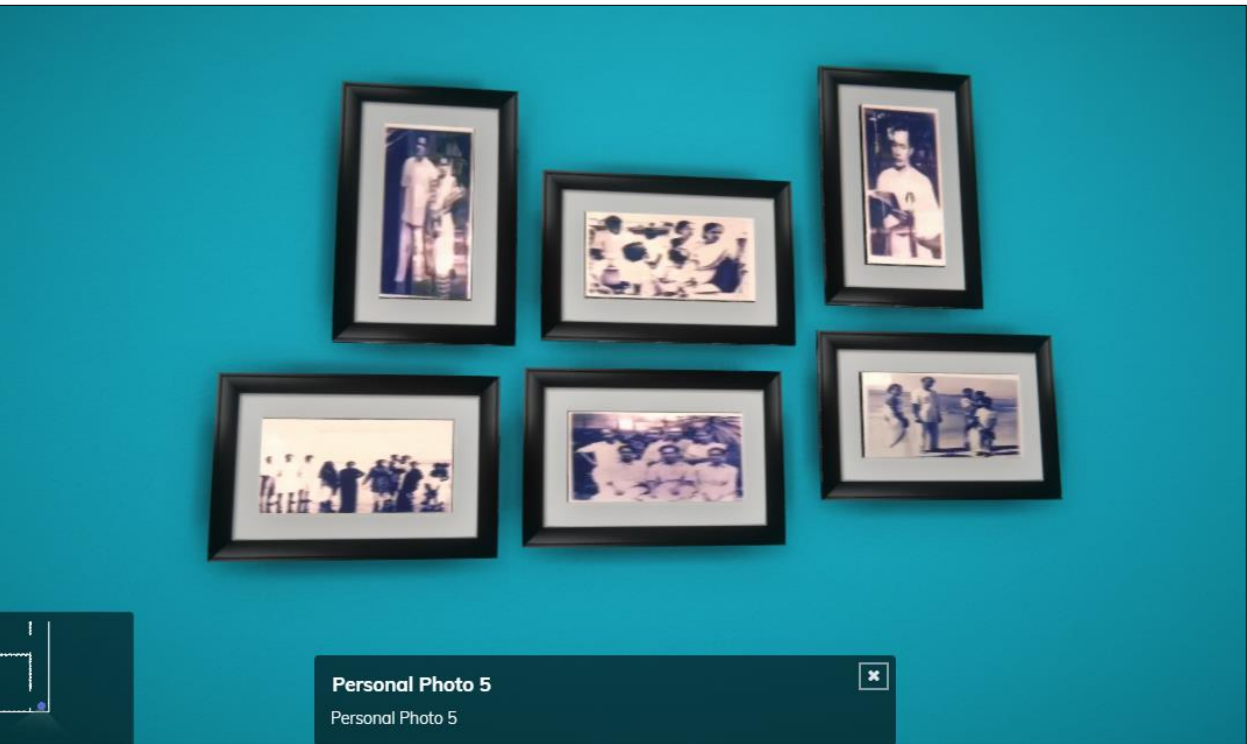


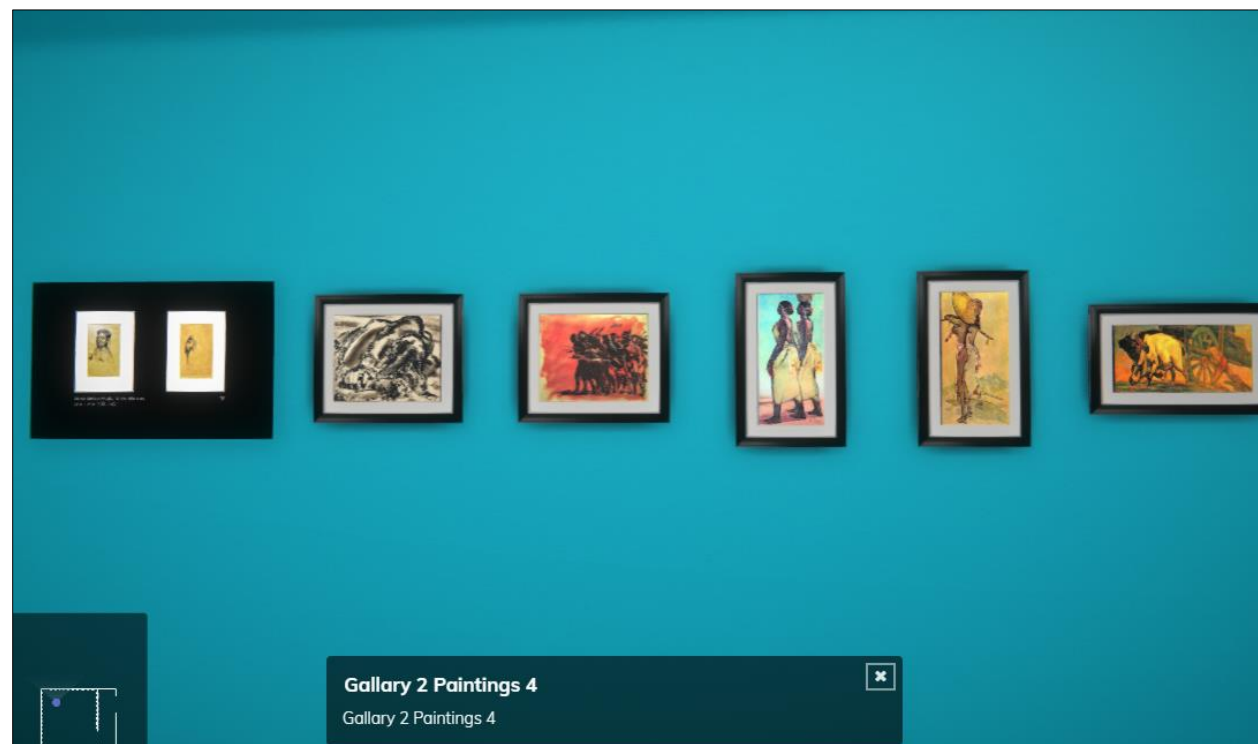
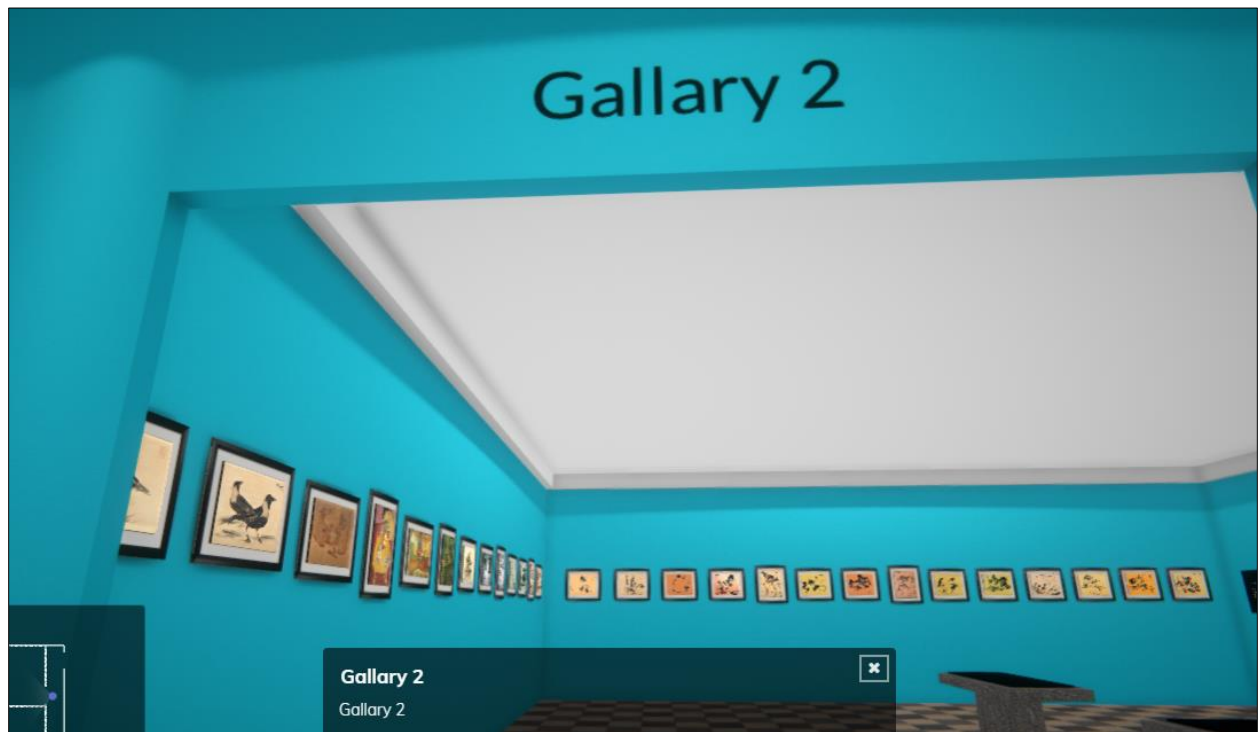
8.2 3D Exhibition

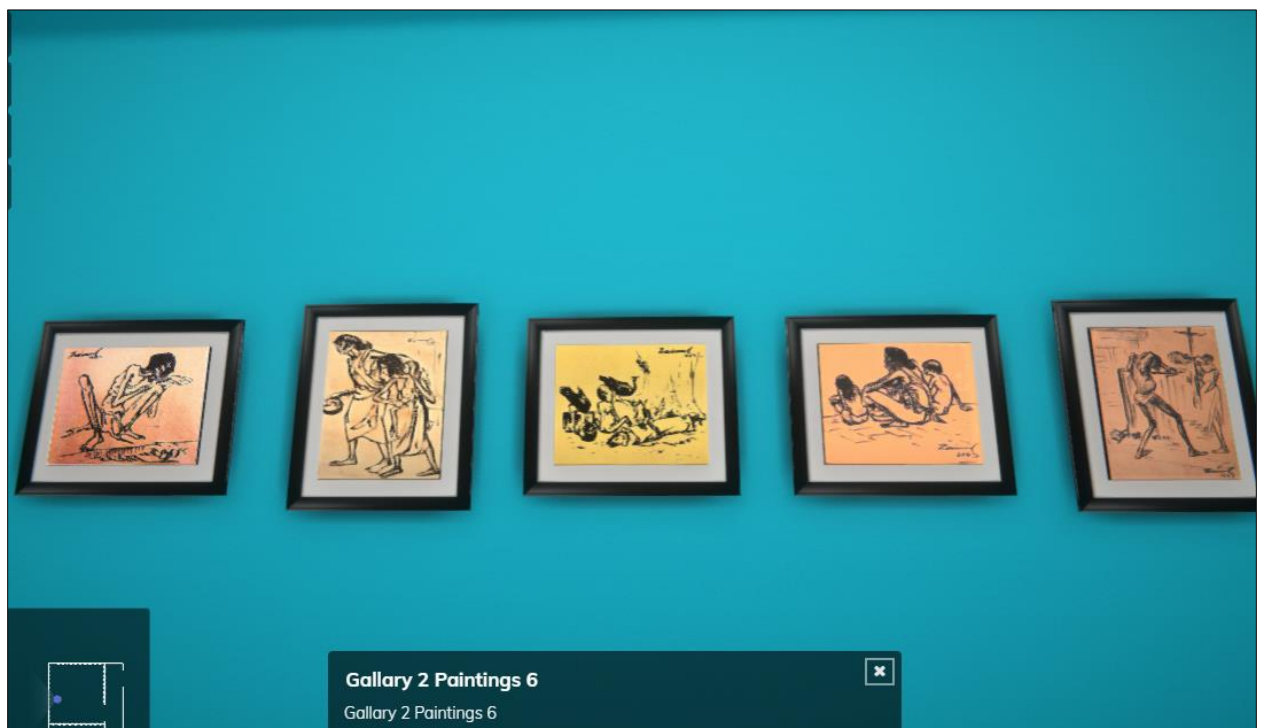
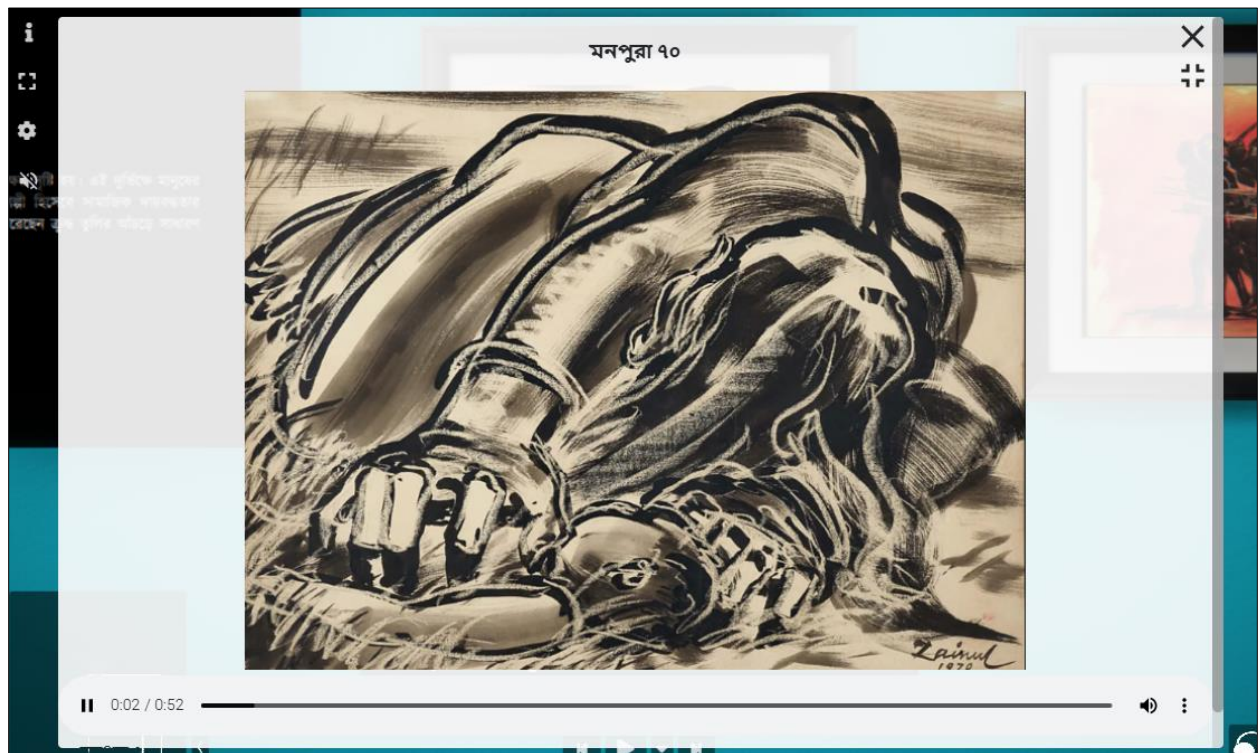


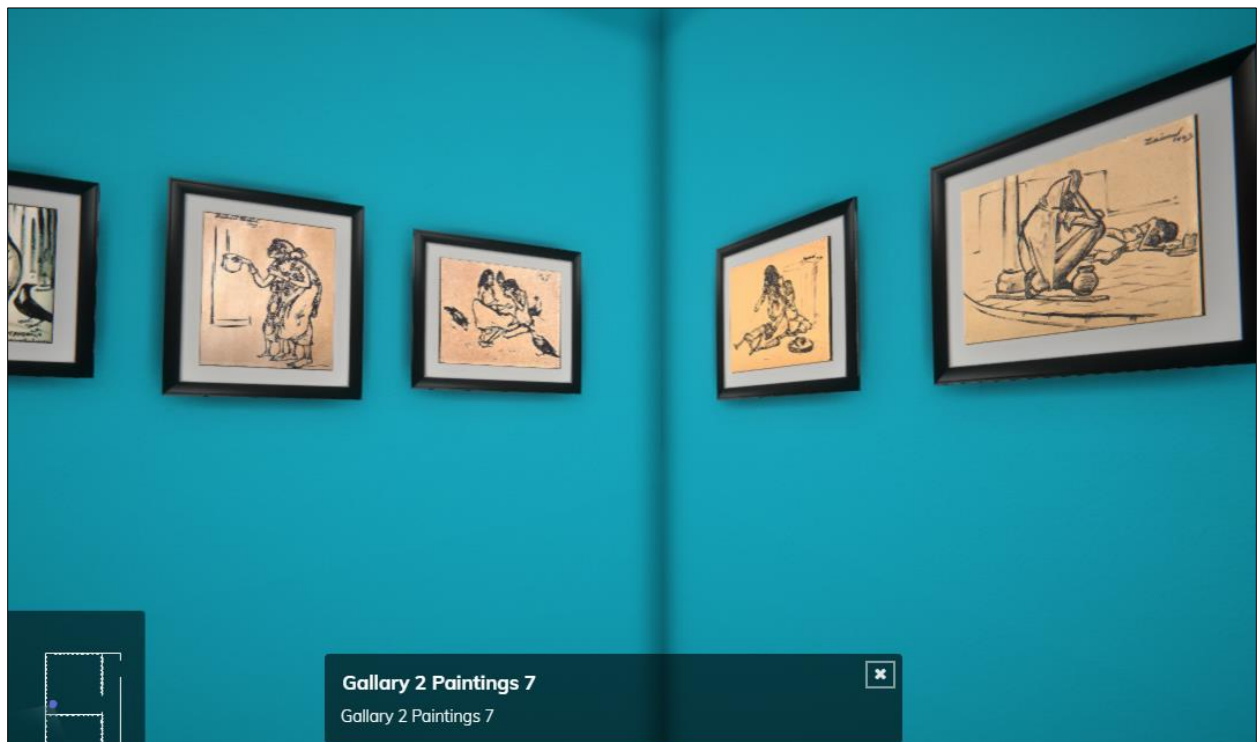
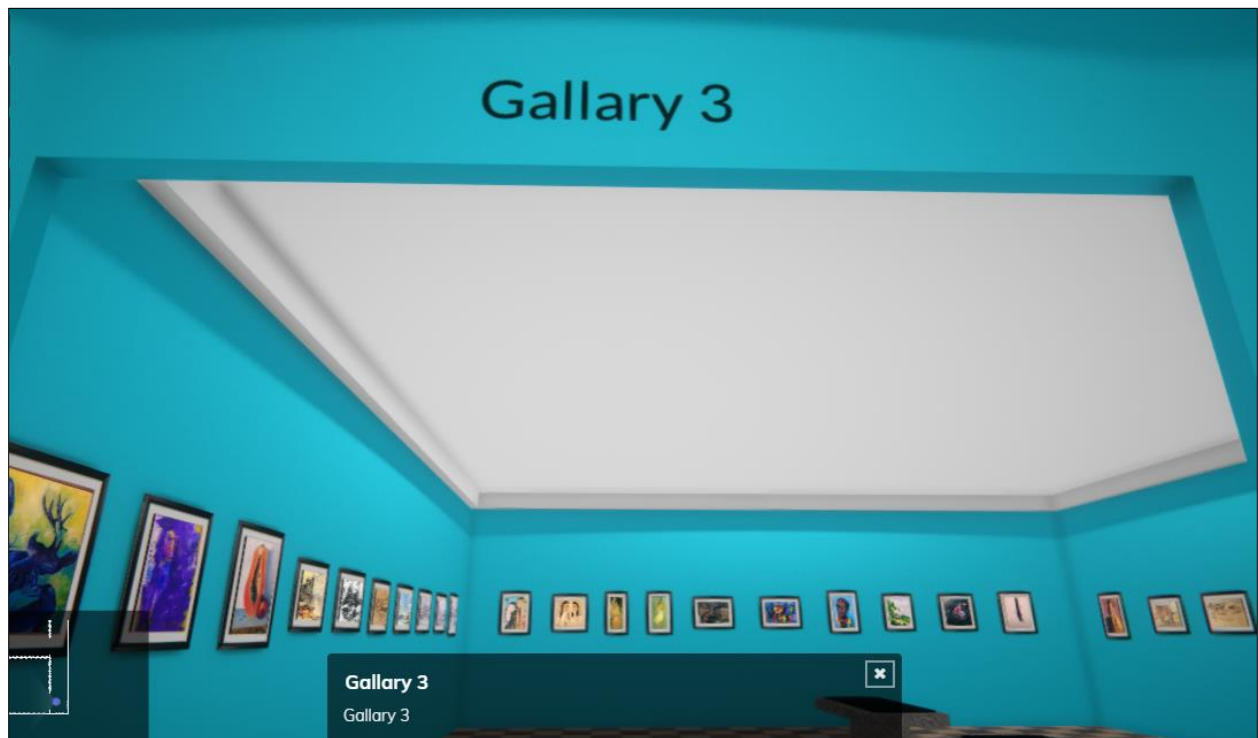












9. Advanced Requirement Management

This section outlines how requirement changes and project risks are managed throughout the development lifecycle to ensure the system remains aligned with stakeholder goals.

9.1 Change Management

Change management ensures that any proposed modifications to requirements are evaluated, approved, and tracked systematically to avoid project disruption.

Change Management Process:

Step	Description
1. Request Initiation	Any stakeholder (e.g., admin, developer, UX team) can submit a change request with details and justification.
2. Impact Analysis	The project team evaluates how the change affects scope, timeline, cost, and existing requirements.
3. Review & Approval	The Change Control Board (CCB), which includes project leads and client representatives, approves, rejects, or defers the change.
4. Update Documentation	Approved changes are incorporated into the SRS and traceability matrix. Version history is maintained.
5. Implementation & Testing	Development, testing, and re-validation of modified features follow standard procedures.
6. Communication	All stakeholders are notified of the change, updated features, and revised documentation.

9.2 Risk Management

Managing risks helps anticipate and mitigate potential threats to project quality, schedule, or success.

Identified Risks & Mitigation Strategies:

Risk ID	Risk Description	Likelihood	Impact	Mitigation Strategy
R-1	VR rendering issues on low-end devices	High	High	Use performance-optimized 3D models and responsive design testing.
R-2	Multilingual content inconsistency	Medium	Medium	Have translations reviewed by native speakers; test both languages simultaneously.
R-3	User data security breach	Low	High	Use encryption, secure authentication (OAuth), and regular vulnerability scans.
R-4	Delay in audio guide production	Medium	Medium	Parallelize development and media production. Allow phased release of audio content.
R-5	Browser compatibility issues (WebXR, A-Frame)	Medium	High	Use progressive enhancement and fallback support. Test on Chrome, Firefox, Edge, Safari.
R-6	Admin errors during content updates	Medium	Medium	Provide training and role-based access with undo/version controls.
R-7	Change in stakeholder expectations mid-project	Medium	High	Use agile methodology and short release cycles for iterative feedback.
R-8	Loss of connectivity during VR experience	High	Medium	Enable session caching, reconnect handling, and content preloading.
R-9	Low user engagement or feedback	Low	Medium	Conduct surveys, usability testing, and content updates based on analytics.
R-10	Licensing or copyright issues with digital assets	Low	High	Use only verified public domain or licensed materials; keep asset logs.

10. Appendix

This appendix includes supplementary information to support the main sections of the Software Requirements Specification (SRS). It provides a glossary of terms, references used during research, and any relevant stakeholder input that helped shape the system requirements.

10.1 Glossary

Term	Definition
VR (Virtual Reality)	An immersive experience where users interact with a computer-generated 3D environment.
WebXR	A web API that enables immersive VR and AR experiences in the browser.
A-Frame	A web framework for building VR experiences using HTML and JavaScript.
Artifact	A 3D model or image of Zainul Abedin's personal belongings (e.g., brushes, tools).
Interactive Exhibit	An exhibit that responds to user actions such as clicks, zooms, or rotations.
Admin Panel	A web interface where authorized users manage museum content and users.
360-Degree View	A feature that allows users to rotate and examine digital objects or artworks from all angles.
Multilingual Support	System ability to present UI and content in more than one language (e.g., Bangla, English).
User Feedback	Input or suggestions submitted by visitors to help improve the virtual museum experience.
RTM (Requirement Traceability Matrix)	A document that links requirements to their corresponding test cases.

10.2 References

IEEE. *IEEE Std 830-1998 IEEE Recommended Practice for Software Requirements Specifications*. IEEE Computer Society, 1998.

10.3 Interview Notes

- **Curators emphasized** the importance of accurate presentation of Zainul Abedin's life and artefacts.
- **Educators requested** downloadable educational materials for use in schools.
- **Students and researchers** wanted audio guides and high-quality visuals for better understanding.
- **Developers highlighted** the need for responsive design and compatibility across devices.
- **Admins requested** easy-to-use interfaces for uploading and editing content.

10.4 Survey Summary

An online survey was conducted with 20 potential users:

Key Insight	Response Rate
Users prefer Bangla as the primary language	68%
Interest in mobile access	82%
Desire for downloadable educational content	61%
Preference for 360° interaction with artworks	73%
Would use the platform for academic purposes	59%